



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Situation Publication - Restrictions

Release 1.2rev0

Národní Dopravní Informační Centrum

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The specification defines format for publishing information about planned and actual closures and traffic restrictions.

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NTIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 Changelog

version 1.0.0

- Initial format and documentation

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.1.0rc

- added feed type to schema and samples
- added OpenLR location to schema

version 1.2

- added urlLink to situationRecord for additional description on the web
- added: feedDescription and safetyRelatedMessage tags
- corrected location of globalNtworkLinear and globalNetworkPoint in the XML tree (as extension)
- corrected type of the networkVersion to the simple string from multilingual string

1.2 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [\[16157-1\]](#)

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [\[16157-3\]](#), *ElaboratedDataPublication* [\[16157-5\]](#), *PredefinedLocationsPublication* [\[16157-3\]](#) etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing information about road actual and planned closures and restrictions.

2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* [16157-3]. Profile uses only small subset of possible structures.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
   ↳instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
   ↳eu/schema/2/2_0 ../schema/schema.xsd">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>DIC</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication xsi:type="SituationPublication" lang="cs">
10     <feedType>x-format:cz-ndic_d2-common-v1.2</feedType>
11     <publicationTime>2023-08-30T09:18:24+02:00</publicationTime>
12     <publicationCreator>
13       <country>cz</country>
14       <nationalIdentifier>NDIC</nationalIdentifier>
15     </publicationCreator>
```

2.2 Event description

Each event is described by the *situation* element, the described situation can evolve in time, so the element is versioned and has its unique identifier.

The *situationRecord* element can occur more than once in the *situation* element (but each time of a different type), it is usually information related to one event, situation. The *situationRecord* element can evolve over time, so the element is versioned and has its unique identifier derived from the identifier of the parent situation.

The traffic situation publication in particular includes:

- Road manager actions
 - Road works
 - Road management
 - * Road closures
 - * Detours

- * Speed limits
- * Change of traffic arrangements

Note: This publication does not include any status information.

Important: This publication is fully included in the d2-common publication where road closures and road works are published along the other information.

Only the structure provided by the *OperatorActions* class is used to describe events. See the W3C XML Schema or [d2ref] as appropriate for the exact definition.

```
24      <situationRecord version="6" xsi:type="ConstructionWorks" id="3bdeb91a-d680-
    ↪ 47e5-ab5c-c3b9b4a13006_ConstructionWorks">
```

```
197      <situationRecord version="6" xsi:type="RoadOrCarriagewayOrLaneManagement" id=
    ↪ "3bdeb91a-d680-47e5-ab5c-c3b9b4a13006_RoadOrCarriagewayOrLaneManagement">
```

The impact of the event is described in the *impact* element, either in the form of expected delay or an estimate of the residual capacity of the road (when closure = 0).

Any additional supplementary information on the event is given by an html link in the *urlLink* element.

```
48      <urlLink>
49        <urlLinkAddress>https://dopravniinfo.cz/te/3bdeb91a-d680-47e5-ab5c-
    ↪ c3b9b4a13006</urlLinkAddress>
50        <urlLinkType>html</urlLinkType>
51      </urlLink>
```

The indication that this is a safety-related event according to the Delegated Regulation 886/2013 is given in the *safetyRelatedMessage* tag.

2.3 Location description

Each *SituationRecord* has its own position description, which may (or may not) differ within a *Situation*. For example, one *SituationRecord* describing a point source problem that resulted in a linear problem (the second *SituationRecord*).

A location tends to be described in a *SituationRecord* by multiple methods, all referring to the same slice of reality.

```
52      <groupOfLocations xsi:type="Linear">
```

TIME VALIDITY

Time validity is stated independently for every single event (*SituationRecord*) described in a situation and is defined by a time interval between two time instants. The class *Validity* uses exclusively an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Setting a week-day and recurring time validity within the specified period, e.g. for recurring intermittent nightly tunnel closure, is not allowed. For such information, splitting to more messages valid in respective times is used.

Used time validity intervals may fall into past, present or future.

```
31 <validity>
32   <validityStatus>definedByValidityTimeSpec</validityStatus>
33   <validityTimeSpecification>
34     <overallStartTime>2023-08-30T07:00:00+02:00</overallStartTime>
35     <overallEndTime>2023-09-01T23:59:00+02:00</overallEndTime>
36   </validityTimeSpecification>
37 </validity>
```


LOCATION DESCRIPTION

When describing locations, nested descriptions of instances of the *Location* class are **always** used, namely the descendants of the *NetworkLocation* class.

The *Linear* class is always used to describe a position, regardless of the fact that it may be a point position, e.g. an accident, expressible by a single coordinate. The reason for this is to unify descriptions between multiple methods. Area type position descriptions and separate point position descriptions are not used.

52

```
<groupOfLocations xsi:type="Linear">
```

Multiple methods can be used to describe a location, depending on their availability in a given location. At least one description must be given.

The following description lists the methods used for location description. For more details, see the *en-ndic_d2-predefined-location-set-v1.1* profile documentation.

4.1 Location description by means of ALERT-C

The method describes the position using ALERT-C codes, class *alertCLinear* of type *AlertCMethod2Linear* is used. The use of this method depends on the presence of an ALERT-C location at a given location.

The *alertCLinear* method uses a primary location (*alertCMethod2PrimaryPointLocation* element), a secondary location (*alertCMethod2SecondaryPointLocation* element) to describe the event. Of course, the method also contains elements allowing to identify a specific location table.

The primary location is, according to the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the convoy or where the driver first enters the problem area.

In order to find out the direction of the situation, the *alertCDirectionCoded* element is used, which takes the value *positive* if the direction of the traffic flow agrees with the coding of the given route.

Important: WARNING, the meaning of this attribute is therefore exactly the opposite of the *direction* attribute in RDS-TMC, which is used to find the secondary location.

The *alertCLinear* element may occur multiple times in the situation description for long events that must be described by more than one ALERT-C path. This is because ALERT-C only allows locations to be chained within a single path.

Note: The *alertCLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). When primary and secondary locations are the same. The direction can be read from the *alertCDirectionCoded* element).

```

53 <alertCLinear xsi:type="AlertCMethod2Linear">
54   <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
55   <alertCLocationTableNumber>25</alertCLocationTableNumber>
56   <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
57   <alertCDirection>
58     <alertCDirectionCoded>negative</alertCDirectionCoded>
59   </alertCDirection>
60   <alertCMethod2PrimaryPointLocation>
61     <alertCLocation>
62       <specificLocation>8973</specificLocation>
63     </alertCLocation>
64   </alertCMethod2PrimaryPointLocation>
65   <alertCMethod2SecondaryPointLocation>
66     <alertCLocation>
67       <specificLocation>8974</specificLocation>
68     </alertCLocation>
69   </alertCMethod2SecondaryPointLocation>
70 </alertCLinear>
71 <linearWithinLinearElement>
72   <directionRelativeOnLinearSection>aligned</
73   directionRelativeOnLinearSection>
74   <linearElement>
75     <roadNumber>10</roadNumber>
76   </linearElement>
77   <fromPoint xsi:type="DistanceFromLinearElementStart">
78     <distanceAlong>106797</distanceAlong>
79   </fromPoint>
80   <toPoint xsi:type="DistanceFromLinearElementStart">
81     <distanceAlong>107507</distanceAlong>
82   </toPoint>
</linearWithinLinearElement>

```

4.2 Location description using mileage

The method uses *linearWithinLinearElement* for the description of the position using mileage. The *roadNumber* element describes the number of the affected road, the *directionRelativeOnLinearSection* element describes the direction of the affected road, and the *fromPoint* and *toPoint* elements describe the distance of the start and end of the event from the start of the road.

Note: The *linearWithinLinearElement* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short section). When the offsets *fromPoint* and *toPoint* refer to a segment with zero length.

```

71 <linearWithinLinearElement>
72   <directionRelativeOnLinearSection>aligned</
73   directionRelativeOnLinearSection>
74   <linearElement>
75     <roadNumber>10</roadNumber>
76   </linearElement>
77   <fromPoint xsi:type="DistanceFromLinearElementStart">
78     <distanceAlong>106797</distanceAlong>
79   </fromPoint>
80   <toPoint xsi:type="DistanceFromLinearElementStart">

```

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```

80     <distanceAlong>107507</distanceAlong>
81   </toPoint>
82 </linearWithinLinearElement>

```

4.3 Location description using the Global Network extension

Native method of position input in NDIC using **Extension of DATEX II model type B**.

The method describes, in the *globalNetworkLinear* element, the location using the segments *linearWithinLinearGNElement* of the Global Network (GN) road network, identified by the *networkVersion* version. Each segment contains a sequence, an identifier referring to the GN, and a relative truncation (for the first and last segment). Furthermore, the position is bounded by the elements *startPoint* and *endPoint* in coordinates (here S-JTSK).

Note: The *globalNetworkLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). The *startPoint* and *endPoint* positions contain the same coordinate.

```

118 <globalNetworkLinear>
119   <networkVersion>22.12</networkVersion>
120   <linearGeometryType>continuous</linearGeometryType>
121   <startPoint>
122     <sjtskPointCoordinates>
123       <sjtskX>-668325.9586937</sjtskX>
124       <sjtskY>-976774.250129909</sjtskY>
125     </sjtskPointCoordinates>
126   </startPoint>
127   <endPoint>
128     <sjtskPointCoordinates>
129       <sjtskX>-668050.419197776</sjtskX>
130       <sjtskY>-976823.636025029</sjtskY>
131     </sjtskPointCoordinates>
132   </endPoint>
133   <linearWithinLinearGNElement>
134     <sectionId>126853</sectionId>
135     <directionRelativeOnLinearSection>opposite</
136   <directionRelativeOnLinearSection>
137     <orderOfSection>0</orderOfSection>
138     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
139       <percentageDistanceAlong>0.0922403254250082</
140   <percentageDistanceAlong>
141     </fromPoint>
142     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
143       <percentageDistanceAlong>0</percentageDistanceAlong>
144     </toPoint>
145   </linearWithinLinearGNElement>
146   <linearWithinLinearGNElement>
147     <sectionId>126599</sectionId>
148     <directionRelativeOnLinearSection>opposite</
149   <directionRelativeOnLinearSection>
150     <orderOfSection>1</orderOfSection>
151     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
152       <percentageDistanceAlong>1</percentageDistanceAlong>
153     </fromPoint>
154     <toPoint xsi:type="PercentageDistanceAlongLinearElement">

```

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```

152         <percentageDistanceAlong>0</percentageDistanceAlong>
153     </toPoint>
154 </linearWithinLinearGNElement>
155 <linearWithinLinearGNElement>
156     <sectionId>1461717</sectionId>
157     <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
158     <orderOfSection>2</orderOfSection>
159     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
160         <percentageDistanceAlong>1</percentageDistanceAlong>
161     </fromPoint>
162     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
163         <percentageDistanceAlong>0</percentageDistanceAlong>
164     </toPoint>
165 </linearWithinLinearGNElement>
166 <linearWithinLinearGNElement>
167     <sectionId>1461719</sectionId>
168     <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
169     <orderOfSection>3</orderOfSection>
170     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
171         <percentageDistanceAlong>1</percentageDistanceAlong>
172     </fromPoint>
173     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
174         <percentageDistanceAlong>0</percentageDistanceAlong>
175     </toPoint>
176 </linearWithinLinearGNElement>
177 <linearWithinLinearGNElement>
178     <sectionId>2901326</sectionId>
179     <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
180     <orderOfSection>4</orderOfSection>
181     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
182         <percentageDistanceAlong>1</percentageDistanceAlong>
183     </fromPoint>
184     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
185         <percentageDistanceAlong>0.147949455079654</percentageDistanceAlong>
186     </toPoint>
187 </linearWithinLinearGNElement>
188 </globalNetworkLinear>

```

4.4 Location description using OpenLR extension

The OpenLR method in the *openlrExtendedLinear* element, uses a routing algorithm to describe the location and thus allows on-the-fly localization.

The method uses the reference points *openlrLocationReferencePoint* and *openlrLastLocationReferencePoint* containing coordinates, road direction, and a road type. These reference points are used as the start and end to find the road which is further refined using offsets to the actual extent of the event. In case 2 points are not enough the method allows to use intermediate points.

Note: To describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment) the *openlrExtendedLinear* method is also used, when offsets refer to a segment with zero length.

```

84 <openlrExtendedLinear>
85   <firstDirection>
86     <openlrLocationReferencePoint>
87       <openlrCoordinate>
88         <latitude>50.765282088427412</latitude>
89         <longitude>15.337521544012871</longitude>
90       </openlrCoordinate>
91       <openlrLineAttributes>
92         <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
93         <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
94         <openlrBearing>112</openlrBearing>
95       </openlrLineAttributes>
96       <openlrPathAttributes>
97         <openlrLowestFRCToNextLRPoint>FRC1</openlrLowestFRCToNextLRPoint>
98         <openlrDistanceToNextLRPoint>1240</openlrDistanceToNextLRPoint>
99       </openlrPathAttributes>
100     </openlrLocationReferencePoint>
101     <openlrLastLocationReferencePoint>
102       <openlrCoordinate>
103         <latitude>50.763208691718496</latitude>
104         <longitude>15.347566415800618</longitude>
105       </openlrCoordinate>
106       <openlrLineAttributes>
107         <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
108         <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
109         <openlrBearing>283</openlrBearing>
110       </openlrLineAttributes>
111     </openlrLastLocationReferencePoint>
112     <openlrOffsets>
113       <openlrPositiveOffset>120</openlrPositiveOffset>
114       <openlrNegativeOffset>418</openlrNegativeOffset>
115     </openlrOffsets>
116   </firstDirection>
117 </openlrExtendedLinear>

```

APPENDIX A: MESSAGE EXAMPLES

5.1 restriction.xml - road works, single lane alternate traffic

Publication contains exactly one message about road works with single lane alternate traffic.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="SituationPublication" lang="cs">
    <feedType>x-format:cz-ndic_d2-common-v1.2</feedType>
    <publicationTime>2023-08-30T09:18:24+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="3bdeb91a-d680-47e5-ab5c-c3b9b4a13006" version="6">
      <situationVersionTime>2023-08-30T07:00:02+02:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>normalUrgency</urgency>
      </headerInformation>
      <!-- situation record 1 -->
      <situationRecord version="6" xsi:type="ConstructionWorks" id="3bdeb91a-d680-
47e5-ab5c-c3b9b4a13006_ConstructionWorks">
        <situationRecordCreationTime>2023-08-30T07:00:04+02:00</
situationRecordCreationTime>
        <situationRecordVersionTime>2023-08-30T07:00:02+02:00</
situationRecordVersionTime>
        <probabilityOfOccurrence>riskOf</probabilityOfOccurrence>
        <source>
          <sourceIdentification>RSD</sourceIdentification>
        </source>
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2023-08-30T07:00:00+02:00</overallStartTime>
            <overallEndTime>2023-09-01T23:59:00+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
      </situationRecord>
    </situation>
  </payloadPublication>
</d2LogicalModel>
```

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```

    </validityTimeSpecification>
  </validity>
  <generalPublicComment>
    <comment>
      <values>
        <value lang="cs">silnice I/10 (ulice Krkonošská), Desná, okr. Jablonec
↳ nad Nisou, stavební práce, kyvadlový provoz jedním jízdním pruhem, Od 30.08.2023
↳ 07:00 Do 01.09.2023 23:59, Z důvodu provádění stavebních prací bude na dotčeném
↳ úseku silnice I/10 provedena
částečná uzavírka provozu, která bude označena dle schváleného dopravně
inženýrského opatření. V průběhu provádění stavebních prací bude doprava řízena
↳ kyvadlově
dvoucestným signalizačním zařízením., Vydal: Krajský úřad Libereckého kraje</value>
      </values>
    </comment>
  </generalPublicComment>
  <urlLink>
    <urlLinkAddress>https://dopravniinfo.cz/te/3bdeb91a-d680-47e5-ab5c-
↳ c3b9b4a13006</urlLinkAddress>
    <urlLinkType>html</urlLinkType>
  </urlLink>
  <groupOfLocations xsi:type="Linear">
    <alertCLinear xsi:type="AlertCMethod2Linear">
      <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
      <alertCLocationTableNumber>25</alertCLocationTableNumber>
      <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
      <alertCDirection>
        <alertCDirectionCoded>negative</alertCDirectionCoded>
      </alertCDirection>
      <alertCMethod2PrimaryPointLocation>
        <alertCLocation>
          <specificLocation>8973</specificLocation>
        </alertCLocation>
      </alertCMethod2PrimaryPointLocation>
      <alertCMethod2SecondaryPointLocation>
        <alertCLocation>
          <specificLocation>8974</specificLocation>
        </alertCLocation>
      </alertCMethod2SecondaryPointLocation>
    </alertCLinear>
    <linearWithinLinearElement>
      <directionRelativeOnLinearSection>aligned</
↳ directionRelativeOnLinearSection>
      <linearElement>
        <roadNumber>10</roadNumber>
      </linearElement>
      <fromPoint xsi:type="DistanceFromLinearElementStart">
        <distanceAlong>106797</distanceAlong>
      </fromPoint>
      <toPoint xsi:type="DistanceFromLinearElementStart">
        <distanceAlong>107507</distanceAlong>
      </toPoint>
    </linearWithinLinearElement>
    <linearExtension>
      <openlrExtendedLinear>
        <firstDirection>

```

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```

<openlrLocationReferencePoint>
  <openlrCoordinate>
    <latitude>50.765282088427412</latitude>
    <longitude>15.337521544012871</longitude>
  </openlrCoordinate>
  <openlrLineAttributes>
    <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
    <openlrBearing>112</openlrBearing>
  </openlrLineAttributes>
  <openlrPathAttributes>
    <openlrLowestFRCToNextLRPoint>FRC1</openlrLowestFRCToNextLRPoint>
    <openlrDistanceToNextLRPoint>1240</openlrDistanceToNextLRPoint>
  </openlrPathAttributes>
</openlrLocationReferencePoint>
<openlrLastLocationReferencePoint>
  <openlrCoordinate>
    <latitude>50.763208691718496</latitude>
    <longitude>15.347566415800618</longitude>
  </openlrCoordinate>
  <openlrLineAttributes>
    <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
    <openlrBearing>283</openlrBearing>
  </openlrLineAttributes>
</openlrLastLocationReferencePoint>
<openlrOffsets>
  <openlrPositiveOffset>120</openlrPositiveOffset>
  <openlrNegativeOffset>418</openlrNegativeOffset>
</openlrOffsets>
</firstDirection>
</openlrExtendedLinear>
<globalNetworkLinear>
  <networkVersion>22.12</networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <startPoint>
    <sjtskPointCoordinates>
      <sjtskX>-668325.9586937</sjtskX>
      <sjtskY>-976774.250129909</sjtskY>
    </sjtskPointCoordinates>
  </startPoint>
  <endPoint>
    <sjtskPointCoordinates>
      <sjtskX>-668050.419197776</sjtskX>
      <sjtskY>-976823.636025029</sjtskY>
    </sjtskPointCoordinates>
  </endPoint>
  <linearWithinLinearGNElement>
    <sectionId>126853</sectionId>
    <directionRelativeOnLinearSection>opposite</
->directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0.0922403254250082</
->percentageDistanceAlong>
    </fromPoint>

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```

        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
    </linearWithinLinearGNElement>
    <sectionId>126599</sectionId>
    <directionRelativeOnLinearSection>opposite</
    directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</linearWithinLinearGNElement>
  <sectionId>1461717</sectionId>
  <directionRelativeOnLinearSection>opposite</
  directionRelativeOnLinearSection>
  <orderOfSection>2</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>0</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
</linearWithinLinearGNElement>
  <sectionId>1461719</sectionId>
  <directionRelativeOnLinearSection>opposite</
  directionRelativeOnLinearSection>
  <orderOfSection>3</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>0</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
</linearWithinLinearGNElement>
  <sectionId>2901326</sectionId>
  <directionRelativeOnLinearSection>opposite</
  directionRelativeOnLinearSection>
  <orderOfSection>4</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>0.147949455079654</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</groupOfLocations>
<subjects>

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```

    <subjectTypeOfWorks>other</subjectTypeOfWorks>
  </subjects>
  <constructionWorkType>constructionWork</constructionWorkType>
</situationRecord>
<!-- situation record 2 -->
  <situationRecord version="6" xsi:type="RoadOrCarriagewayOrLaneManagement" id=
↪ "3bdeb91a-d680-47e5-ab5c-c3b9b4a13006_RoadOrCarriagewayOrLaneManagement">
    <situationRecordCreationTime>2023-08-30T07:00:04+02:00</
↪ situationRecordCreationTime>
    <situationRecordVersionTime>2023-08-30T07:00:02+02:00</
↪ situationRecordVersionTime>
    <probabilityOfOccurrence>riskOf</probabilityOfOccurrence>
    <source>
      <sourceIdentification>RSD</sourceIdentification>
    </source>
    <validity>
      <validityStatus>definedByValidityTimeSpec</validityStatus>
      <validityTimeSpecification>
        <overallStartTime>2023-08-30T07:00:00+02:00</overallStartTime>
        <overallEndTime>2023-09-01T23:59:00+02:00</overallEndTime>
      </validityTimeSpecification>
    </validity>
    <generalPublicComment>
      <comment>
        <values>
          <value lang="cs">silnice I/10 (ulice Krkonošská), Desná, okr. Jablonec,
↪ nad Nisou, stavební práce, kyvadlový provoz jedním jízdním pruhem, Od 30.08.2023,
↪ 07:00 Do 01.09.2023 23:59, Z důvodu provádění stavebních prací bude na dotčeném,
↪ úseku silnice I/10 provedena
částečná uzavírka provozu, která bude označena dle schváleného dopravně
inženýrského opatření. V průběhu provádění stavebních prací bude doprava řízena,
↪ kyvadlově
dvoucestným signalizačním zařízením., Vydal: Krajský úřad Libereckého kraje</value>
        </values>
      </comment>
    </generalPublicComment>
    <urlLink>
      <urlLinkAddress>https://dopravniinfo.cz/te/3bdeb91a-d680-47e5-ab5c-
↪ c3b9b4a13006</urlLinkAddress>
      <urlLinkType>html</urlLinkType>
    </urlLink>
    <groupOfLocations xsi:type="Linear">
      <alertCLinear xsi:type="AlertCMethod2Linear">
        <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
        <alertCLocationTableNumber>25</alertCLocationTableNumber>
        <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
        <alertCDirection>
          <alertCDirectionCoded>negative</alertCDirectionCoded>
        </alertCDirection>
        <alertCMethod2PrimaryPointLocation>
          <alertCLocation>
            <specificLocation>8973</specificLocation>
          </alertCLocation>
        </alertCMethod2PrimaryPointLocation>
        <alertCMethod2SecondaryPointLocation>
          <alertCLocation>

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```

        <specificLocation>8974</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
<linearWithinLinearElement>
  <directionRelativeOnLinearSection>aligned</
->directionRelativeOnLinearSection>
  <linearElement>
    <roadNumber>10</roadNumber>
  </linearElement>
  <fromPoint xsi:type="DistanceFromLinearElementStart">
    <distanceAlong>106797</distanceAlong>
  </fromPoint>
  <toPoint xsi:type="DistanceFromLinearElementStart">
    <distanceAlong>107507</distanceAlong>
  </toPoint>
</linearWithinLinearElement>
<linearExtension>
  <openlrExtendedLinear>
    <firstDirection>
      <openlrLocationReferencePoint>
        <openlrCoordinate>
          <latitude>50.765282088427412</latitude>
          <longitude>15.337521544012871</longitude>
        </openlrCoordinate>
        <openlrLineAttributes>
          <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
          <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
          <openlrBearing>112</openlrBearing>
        </openlrLineAttributes>
        <openlrPathAttributes>
          <openlrLowestFRCToNextLRPoint>FRC1</openlrLowestFRCToNextLRPoint>
          <openlrDistanceToNextLRPoint>1240</openlrDistanceToNextLRPoint>
        </openlrPathAttributes>
      </openlrLocationReferencePoint>
      <openlrLastLocationReferencePoint>
        <openlrCoordinate>
          <latitude>50.763208691718496</latitude>
          <longitude>15.347566415800618</longitude>
        </openlrCoordinate>
        <openlrLineAttributes>
          <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
          <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
          <openlrBearing>283</openlrBearing>
        </openlrLineAttributes>
      </openlrLastLocationReferencePoint>
      <openlrOffsets>
        <openlrPositiveOffset>120</openlrPositiveOffset>
        <openlrNegativeOffset>418</openlrNegativeOffset>
      </openlrOffsets>
    </firstDirection>
  </openlrExtendedLinear>
</globalNetworkLinear>
  <networkVersion>22.12</networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <startPoint>

```

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```

        <sjtskPointCoordinates>
          <sjtskX>-668325.9586937</sjtskX>
          <sjtskY>-976774.250129909</sjtskY>
        </sjtskPointCoordinates>
      </startPoint>
      <endPoint>
        <sjtskPointCoordinates>
          <sjtskX>-668050.419197776</sjtskX>
          <sjtskY>-976823.636025029</sjtskY>
        </sjtskPointCoordinates>
      </endPoint>
      <linearWithinLinearGNElement>
        <sectionId>126853</sectionId>
        <directionRelativeOnLinearSection>opposite</
    ↪ directionRelativeOnLinearSection>
        <orderOfSection>0</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0.0922403254250082</
    ↪ percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>126599</sectionId>
        <directionRelativeOnLinearSection>opposite</
    ↪ directionRelativeOnLinearSection>
        <orderOfSection>1</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>1461717</sectionId>
        <directionRelativeOnLinearSection>opposite</
    ↪ directionRelativeOnLinearSection>
        <orderOfSection>2</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>1461719</sectionId>
        <directionRelativeOnLinearSection>opposite</
    ↪ directionRelativeOnLinearSection>
        <orderOfSection>3</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>

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```

        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>2901326</sectionId>
        <directionRelativeOnLinearSection>opposite</
→ directionRelativeOnLinearSection>
        <orderOfSection>4</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0.147949455079654</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
    </globalNetworkLinear>
  </linearExtension>
</groupOfLocations>
  <complianceOption>mandatory</complianceOption>
  <roadOrCarriagewayOrLaneManagementType>singleAlternateLineTraffic</
→ roadOrCarriagewayOrLaneManagementType>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication (file) as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of one specific situation

The following element *situation* may be repeated and describes individual separate traffic information. The *situationRecord* element occurs exactly twice in the *situation* element in this case.

The *situationVersionTime* element indicates the time of release of this version of the situation as a whole.

The *headerInformation* element describes how the described information is handled.

The *probabilityOfOccurrence* element with value *riskOf* refers to a planned closure that has not yet occurred.

The *urlLink* element contains a link to a web site where more information about the situation is available.

Event description (what happened)

The first element *situationRecord* of type *xsi:type="ConstructionWorks"* contains the description of the cause, i.e. the ongoing construction work *constructionWork*. The second element *situationRecord* of type *xsi:type="RoadOrCarriagewayOrLaneManagement"* contains a description of the impact, i.e. alternating one-way traffic *singleAlternateLineTraffic*.

Both elements of *situationRecord* are children of the *OperatorActions* class used in this format.

A *complianceOption* with value *mandatory* specifies that this is a binding constraint.

Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

Event location

The location of the event described directly in the *groupOfLocations* element by the nested description is of the *xsi:type="Linear"* type, i.e. the linear location type.

In the example, 4 location descriptions are used:

- tmc location using location tables in the *alertCLinear* element of type *AlertCMethod2Linear*,
- description using stationing in the *linearWithinLinearElement* element,
- description by reference to Global Network section ids in the *globalNetworkLinear* extension, and
- description using the OpenLR routing method in the *openlrExtendedLinear* extension.

All location description methods used express the same location.

Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

The *generalPublicComment* element contains a general description of the event described in the message, usually only in italics. It is a free description of the situation by its creator.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:element name="globalNetworkLinear" type="D2LogicalModel:GlobalNetworkLinear"
  ↳ minOccurs="0" />
18      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
19    </xs:sequence>
20  </xs:complexType>
21  <xs:complexType name="_PointExtensionType">
22    <xs:sequence>
23      <xs:element name="pointWithinLinearGNElement" type=
  ↳ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
24      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint"
  ↳ minOccurs="0" />
25      <xs:element name="pointByMultiCoordinates" type=
  ↳ "D2LogicalModel:PointByMultiCoordinates" minOccurs="0" />
26      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
27    </xs:sequence>
28  </xs:complexType>

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```

29 <xs:complexType name="_SituationRecordExtensionType">
30   <xs:sequence>
31     <xs:element name="situationRecordExtendedApproved" type=
↪ "D2LogicalModel:SituationRecordExtendedApproved" minOccurs="0" />
32     <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
↪ "unbounded" />
33   </xs:sequence>
34 </xs:complexType>
35 <xs:complexType name="AffectedCarriagewayAndLanes">
36   <xs:sequence>
37     <xs:element name="carriageway" type="D2LogicalModel:CarriagewayEnum" minOccurs=
↪ "1" maxOccurs="1" />
38     <xs:element name="lane" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs=
↪ "unbounded" />
39     <xs:element name="footpath" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
40     <xs:element name="lengthAffected" type="D2LogicalModel:MetresAsFloat" minOccurs=
↪ "0" maxOccurs="1" />
41     <xs:element name="affectedCarriagewayAndLanesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
42   </xs:sequence>
43 </xs:complexType>
44 <xs:complexType name="AlertCDirection">
45   <xs:sequence>
46     <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
↪ " minOccurs="1" maxOccurs="1" />
47     <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
↪ " minOccurs="0" maxOccurs="1" />
48     <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
↪ "0" maxOccurs="1" />
49     <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
50   </xs:sequence>
51 </xs:complexType>
52 <xs:simpleType name="AlertCDirectionEnum">
53   <xs:restriction base="xs:string">
54     <xs:enumeration value="both" />
55     <xs:enumeration value="negative" />
56     <xs:enumeration value="positive" />
57     <xs:enumeration value="unknown" />
58   </xs:restriction>
59 </xs:simpleType>
60 <xs:complexType name="AlertCLinear" abstract="true">
61   <xs:sequence>
62     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
63     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
64     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
65     <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
66   </xs:sequence>
67 </xs:complexType>
68 <xs:complexType name="AlertCLinearByCode">
69   <xs:complexContent>

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```

70     <xs:extension base="D2LogicalModel:AlertCLinear">
71         <xs:sequence>
72             <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
73             <xs:element name="locationCodeForLinearLocation" type=
74 ↪ "D2LogicalModel:AlertCLocation" />
75             <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
76 ↪ ExtensionType" minOccurs="0" />
77         </xs:sequence>
78     </xs:extension>
79 </xs:complexType>
80 <xs:complexType name="AlertCLocation">
81     <xs:sequence>
82         <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
83 ↪ minOccurs="0" maxOccurs="1" />
84         <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
85 ↪ minOccurs="1" maxOccurs="1" />
86         <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
87 ↪ minOccurs="0" />
88     </xs:sequence>
89 </xs:complexType>
90 <xs:simpleType name="AlertCLocationCode">
91     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
92 </xs:simpleType>
93 <xs:complexType name="AlertCMethod2Linear">
94     <xs:complexContent>
95         <xs:extension base="D2LogicalModel:AlertCLinear">
96             <xs:sequence>
97                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
98                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
99 ↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
100                 <xs:element name="alertCMethod2SecondaryPointLocation" type=
101 ↪ "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
102                 <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
103 ↪ ExtensionType" minOccurs="0" />
104             </xs:sequence>
105         </xs:extension>
106     </xs:complexContent>
107 </xs:complexType>
108 <xs:complexType name="AlertCMethod2Point">
109     <xs:complexContent>
110         <xs:extension base="D2LogicalModel:AlertCPoint">
111             <xs:sequence>
112                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
113                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
114 ↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
115                 <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
116 ↪ ExtensionType" minOccurs="0" />
117             </xs:sequence>
118         </xs:extension>
119     </xs:complexContent>
120 </xs:complexType>
121 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
122     <xs:sequence>
123         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
124         <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=

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```

116   <xs:sequence>
117 </xs:complexType>
118 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
119   <xs:sequence>
120     <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
121     <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
122 <xs:sequence>
123 </xs:complexType>
124 <xs:complexType name="AlertCPoint" abstract="true">
125   <xs:sequence>
126     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
127 <minOccurs="1" maxOccurs="1" />
128     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
129 <minOccurs="1" maxOccurs="1" />
130     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
131 <minOccurs="1" maxOccurs="1" />
132     <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
133 <minOccurs="0" />
134   </xs:sequence>
135 </xs:complexType>
136 <xs:simpleType name="AngleInDegrees">
137   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
138 </xs:simpleType>
139 <xs:complexType name="Area">
140   <xs:complexContent>
141     <xs:extension base="D2LogicalModel:Location">
142       <xs:sequence>
143         <xs:element name="areaExtension" type="D2LogicalModel:_ExtensionType"
144 <minOccurs="0" />
145       </xs:sequence>
146     </xs:extension>
147   </xs:complexContent>
148 </xs:complexType>
149 <xs:complexType name="AreaDestination">
150   <xs:complexContent>
151     <xs:extension base="D2LogicalModel:Destination">
152       <xs:sequence>
153         <xs:element name="area" type="D2LogicalModel:Area" />
154         <xs:element name="areaDestinationExtension" type="D2LogicalModel:_
155 <ExtensionType" minOccurs="0" />
156       </xs:sequence>
157     </xs:extension>
158   </xs:complexContent>
159 </xs:complexType>
160 <xs:complexType name="AxleFlowValue">
161   <xs:complexContent>
    <xs:extension base="D2LogicalModel:DataValue">
      <xs:sequence>
        <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour"
        <minOccurs="1" maxOccurs="1" />
        <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
        <ExtensionType" minOccurs="0" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>

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```

162     </xs:complexContent>
163 </xs:complexType>
164 <xs:simpleType name="AxlesPerHour">
165     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
166 </xs:simpleType>
167 <xs:simpleType name="Boolean">
168     <xs:restriction base="xs:boolean" />
169 </xs:simpleType>
170 <xs:simpleType name="CarriagewayEnum">
171     <xs:restriction base="xs:string">
172         <xs:enumeration value="connectingCarriageway" />
173         <xs:enumeration value="entrySlipRoad" />
174         <xs:enumeration value="exitSlipRoad" />
175         <xs:enumeration value="flyover" />
176         <xs:enumeration value="leftHandFeederRoad" />
177         <xs:enumeration value="leftHandParallelCarriageway" />
178         <xs:enumeration value="mainCarriageway" />
179         <xs:enumeration value="oppositeCarriageway" />
180         <xs:enumeration value="parallelCarriageway" />
181         <xs:enumeration value="rightHandFeederRoad" />
182         <xs:enumeration value="rightHandParallelCarriageway" />
183         <xs:enumeration value="roundabout" />
184         <xs:enumeration value="serviceRoad" />
185         <xs:enumeration value="slipRoads" />
186         <xs:enumeration value="underpass" />
187     </xs:restriction>
188 </xs:simpleType>
189 <xs:complexType name="Cause" abstract="true">
190     <xs:sequence>
191         <xs:element name="causeExtension" type="D2LogicalModel:_ExtensionType"
192         ↪ minOccurs="0" />
193     </xs:sequence>
194 </xs:complexType>
195 <xs:simpleType name="CauseTypeEnum">
196     <xs:restriction base="xs:string">
197         <xs:enumeration value="accident" />
198         <xs:enumeration value="congestion" />
199         <xs:enumeration value="earlierAccident" />
200         <xs:enumeration value="earlierEvent" />
201         <xs:enumeration value="earlierIncident" />
202         <xs:enumeration value="equipmentFailure" />
203         <xs:enumeration value="excessiveHeat" />
204         <xs:enumeration value="frost" />
205         <xs:enumeration value="holidayTraffic" />
206         <xs:enumeration value="infrastructureFailure" />
207         <xs:enumeration value="largeNumbersOfVisitors" />
208         <xs:enumeration value="obstruction" />
209         <xs:enumeration value="pollutionAlert" />
210         <xs:enumeration value="poorWeather" />
211         <xs:enumeration value="problemsAtBorderPost" />
212         <xs:enumeration value="problemsAtCustomPost" />
213         <xs:enumeration value="problemsOnLocalRoads" />
214         <xs:enumeration value="radioactiveLeakAlert" />
215         <xs:enumeration value="roadsideEvent" />
216         <xs:enumeration value="rubberNecking" />
217         <xs:enumeration value="securityIncident" />

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217     <xs:enumeration value="shearWeightOfTraffic" />
218     <xs:enumeration value="technicalProblems" />
219     <xs:enumeration value="terrorism" />
220     <xs:enumeration value="toxicCloudAlert" />
221     <xs:enumeration value="vandalism" />
222     <xs:enumeration value="other" />
223   </xs:restriction>
224 </xs:simpleType>
225 <xs:complexType name="Comment">
226   <xs:sequence>
227     <xs:element name="comment" type="D2LogicalModel:MultilingualString" minOccurs="1
228     ↪ " maxOccurs="1" />
229     <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs="0"
230     ↪ maxOccurs="1" />
231     <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum" minOccurs=
232     ↪ "0" maxOccurs="1" />
233     <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"
234     ↪ minOccurs="0" />
235   </xs:sequence>
236 </xs:complexType>
237 <xs:simpleType name="CommentTypeEnum">
238   <xs:restriction base="xs:string">
239     <xs:enumeration value="abnormalLoadMovementNote" />
240     <xs:enumeration value="dataProcessingNote" />
241     <xs:enumeration value="description" />
242     <xs:enumeration value="internalNote" />
243     <xs:enumeration value="locationDescriptor" />
244     <xs:enumeration value="warning" />
245     <xs:enumeration value="other" />
246   </xs:restriction>
247 </xs:simpleType>
248 <xs:simpleType name="ComparisonOperatorEnum">
249   <xs:restriction base="xs:string">
250     <xs:enumeration value="equalTo" />
251     <xs:enumeration value="greaterThan" />
252     <xs:enumeration value="greaterThanOrEqualTo" />
253     <xs:enumeration value="lessThan" />
254     <xs:enumeration value="lessThanOrEqualTo" />
255   </xs:restriction>
256 </xs:simpleType>
257 <xs:simpleType name="ComplianceOptionEnum">
258   <xs:restriction base="xs:string">
259     <xs:enumeration value="advisory" />
260     <xs:enumeration value="mandatory" />
261   </xs:restriction>
262 </xs:simpleType>
263 <xs:simpleType name="ComputationMethodEnum">
264   <xs:restriction base="xs:string">
265     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
266     ↪ >
267     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
268     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
269     <xs:enumeration value="medianOfSamplesInATimePeriod" />
270     <xs:enumeration value="movingAverageOfSamples" />
271   </xs:restriction>
272 </xs:simpleType>

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268 <xs:complexType name="ConcentrationOfVehiclesValue">
269   <xs:complexContent>
270     <xs:extension base="D2LogicalModel:DataValue">
271       <xs:sequence>
272         <xs:element name="concentrationOfVehicles" type=
↪ "D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1" />
273         <xs:element name="concentrationOfVehiclesValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
274       </xs:sequence>
275     </xs:extension>
276   </xs:complexContent>
277 </xs:complexType>
278 <xs:simpleType name="ConcentrationVehiclesPerKilometre">
279   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
280 </xs:simpleType>
281 <xs:simpleType name="ConfidentialityValueEnum">
282   <xs:restriction base="xs:string">
283     <xs:enumeration value="internalUse" />
284     <xs:enumeration value="noRestriction" />
285     <xs:enumeration value="restrictedToAuthorities" />
286     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
287     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
288     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
289   </xs:restriction>
290 </xs:simpleType>
291 <xs:complexType name="ConstructionWorks">
292   <xs:complexContent>
293     <xs:extension base="D2LogicalModel:Roadworks">
294       <xs:sequence>
295         <xs:element name="constructionWorkType" type=
↪ "D2LogicalModel:ConstructionWorkTypeEnum" minOccurs="0" maxOccurs="1" />
296         <xs:element name="constructionWorksExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
297       </xs:sequence>
298     </xs:extension>
299   </xs:complexContent>
300 </xs:complexType>
301 <xs:simpleType name="ConstructionWorkTypeEnum">
302   <xs:restriction base="xs:string">
303     <xs:enumeration value="blastingWork" />
304     <xs:enumeration value="constructionWork" />
305     <xs:enumeration value="demolitionWork" />
306     <xs:enumeration value="roadImprovementOrUpgrading" />
307     <xs:enumeration value="roadWideningWork" />
308   </xs:restriction>
309 </xs:simpleType>
310 <xs:simpleType name="CountryEnum">
311   <xs:restriction base="xs:string">
312     <xs:enumeration value="at" />
313     <xs:enumeration value="be" />
314     <xs:enumeration value="bg" />
315     <xs:enumeration value="ch" />
316     <xs:enumeration value="cs" />
317     <xs:enumeration value="cy" />
318     <xs:enumeration value="cz" />
319     <xs:enumeration value="de" />

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320     <xs:enumeration value="dk" />
321     <xs:enumeration value="ee" />
322     <xs:enumeration value="es" />
323     <xs:enumeration value="fi" />
324     <xs:enumeration value="fo" />
325     <xs:enumeration value="fr" />
326     <xs:enumeration value="gb" />
327     <xs:enumeration value="gg" />
328     <xs:enumeration value="gi" />
329     <xs:enumeration value="gr" />
330     <xs:enumeration value="hr" />
331     <xs:enumeration value="hu" />
332     <xs:enumeration value="ie" />
333     <xs:enumeration value="im" />
334     <xs:enumeration value="is" />
335     <xs:enumeration value="it" />
336     <xs:enumeration value="je" />
337     <xs:enumeration value="li" />
338     <xs:enumeration value="lt" />
339     <xs:enumeration value="lu" />
340     <xs:enumeration value="lv" />
341     <xs:enumeration value="ma" />
342     <xs:enumeration value="mc" />
343     <xs:enumeration value="mk" />
344     <xs:enumeration value="mt" />
345     <xs:enumeration value="nl" />
346     <xs:enumeration value="no" />
347     <xs:enumeration value="pl" />
348     <xs:enumeration value="pt" />
349     <xs:enumeration value="ro" />
350     <xs:enumeration value="se" />
351     <xs:enumeration value="si" />
352     <xs:enumeration value="sk" />
353     <xs:enumeration value="sm" />
354     <xs:enumeration value="tr" />
355     <xs:enumeration value="va" />
356     <xs:enumeration value="other" />
357 </xs:restriction>
358 </xs:simpleType>
359 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
360   <xs:unique name="_d2LogicalModelSituationRecordConstraint">
361     <xs:selector xpath="."/><xs:selector xpath="D2LogicalModel:situationRecord" />
362     <xs:field xpath="@id" />
363     <xs:field xpath="@version" />
364   </xs:unique>
365   <xs:unique name="_d2LogicalModelSituationConstraint">
366     <xs:selector xpath="."/><xs:selector xpath="D2LogicalModel:situation" />
367     <xs:field xpath="@id" />
368     <xs:field xpath="@version" />
369   </xs:unique>
370 </xs:element>
371 <xs:complexType name="D2LogicalModel">
372   <xs:sequence>
373     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
374     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
↪ minOccurs="0" />

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375     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
376   </xs:sequence>
377   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
378 </xs:complexType>
379 <xs:complexType name="DataValue" abstract="true">
380   <xs:sequence>
381     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
382     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
383     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
384   </xs:sequence>
385   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
386   <xs:attribute name="computationalMethod" type=
↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
387   <xs:attribute name="numberOfIncompleteInputs" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
388   <xs:attribute name="numberOfInputValuesUsed" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
389   <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
390   <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"
↪ />
391   <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
↪ " use="optional" />
392 </xs:complexType>
393 <xs:simpleType name="DateTime">
394   <xs:restriction base="xs:dateTime" />
395 </xs:simpleType>
396 <xs:complexType name="DateTimeValue">
397   <xs:complexContent>
398     <xs:extension base="D2LogicalModel:DataValue">
399       <xs:sequence>
400         <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1"
↪ maxOccurs="1" />
401         <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
402       </xs:sequence>
403     </xs:extension>
404   </xs:complexContent>
405 </xs:complexType>
406 <xs:complexType name="Delays">
407   <xs:sequence>
408     <xs:element name="delaysType" type="D2LogicalModel:DelaysTypeEnum" minOccurs="0
↪ " maxOccurs="1" />
409     <xs:element name="delayTimeValue" type="D2LogicalModel:Seconds" minOccurs="0"
↪ maxOccurs="1" />
410     <xs:element name="delaysExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
411   </xs:sequence>
412 </xs:complexType>
413 <xs:simpleType name="DelaysTypeEnum">
414   <xs:restriction base="xs:string">
415     <xs:enumeration value="delays" />
416     <xs:enumeration value="delaysOfUncertainDuration" />

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417     <xs:enumeration value="longDelays" />
418     <xs:enumeration value="veryLongDelays" />
419   </xs:restriction>
420 </xs:simpleType>
421 <xs:complexType name="Destination" abstract="true">
422   <xs:sequence>
423     <xs:element name="destinationExtension" type="D2LogicalModel:_ExtensionType"
424     ↪ minOccurs="0" />
425   </xs:sequence>
426 </xs:complexType>
427 <xs:simpleType name="DirectionEnum">
428   <xs:restriction base="xs:string">
429     <xs:enumeration value="allDirections" />
430     <xs:enumeration value="bothWays" />
431     <xs:enumeration value="clockwise" />
432     <xs:enumeration value="anticlockwise" />
433     <xs:enumeration value="innerRing" />
434     <xs:enumeration value="outerRing" />
435     <xs:enumeration value="northBound" />
436     <xs:enumeration value="northEastBound" />
437     <xs:enumeration value="eastBound" />
438     <xs:enumeration value="southEastBound" />
439     <xs:enumeration value="southBound" />
440     <xs:enumeration value="southWestBound" />
441     <xs:enumeration value="westBound" />
442     <xs:enumeration value="northWestBound" />
443     <xs:enumeration value="inboundTowardsTown" />
444     <xs:enumeration value="outboundFromTown" />
445     <xs:enumeration value="unknown" />
446     <xs:enumeration value="opposite" />
447     <xs:enumeration value="other" />
448   </xs:restriction>
449 </xs:simpleType>
450 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
451   <xs:sequence>
452     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
453     ↪ ExtensionType" minOccurs="0" />
454   </xs:sequence>
455 </xs:complexType>
456 <xs:complexType name="DistanceFromLinearElementReferent">
457   <xs:complexContent>
458     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
459       <xs:sequence>
460         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat"
461         ↪ minOccurs="1" maxOccurs="1" />
462         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
463         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
464         ↪ "0" />
465         <xs:element name="distanceFromLinearElementReferentExtension" type=
466         ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
467       </xs:sequence>
468     </xs:extension>
469   </xs:complexContent>
470 </xs:complexType>
471 <xs:complexType name="DistanceFromLinearElementStart">
472   <xs:complexContent>

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468     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
469         <xs:sequence>
470             <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat"
↳minOccurs="1" maxOccurs="1" />
471             <xs:element name="distanceFromLinearElementStartExtension" type=
↳"D2LogicalModel:_ExtensionType" minOccurs="0" />
472         </xs:sequence>
473     </xs:extension>
474 </xs:complexContent>
475 </xs:complexType>
476 <xs:complexType name="ElaboratedDataFault">
477     <xs:complexContent>
478         <xs:extension base="D2LogicalModel:Fault">
479             <xs:sequence>
480                 <xs:element name="elaboratedDataFault" type=
↳"D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
481                 <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
↳ExtensionType" minOccurs="0" />
482             </xs:sequence>
483         </xs:extension>
484     </xs:complexContent>
485 </xs:complexType>
486 <xs:simpleType name="ElaboratedDataFaultEnum">
487     <xs:restriction base="xs:string">
488         <xs:enumeration value="intermittentDataValues" />
489         <xs:enumeration value="noDataValuesAvailable" />
490         <xs:enumeration value="spuriousUnreliableDataValues" />
491         <xs:enumeration value="unspecifiedOrUnknownFault" />
492         <xs:enumeration value="other" />
493     </xs:restriction>
494 </xs:simpleType>
495 <xs:complexType name="Exchange">
496     <xs:sequence>
497         <xs:element name="supplierIdentification" type=
↳"D2LogicalModel:InternationalIdentifier" />
498         <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
↳minOccurs="0" />
499     </xs:sequence>
500 </xs:complexType>
501 <xs:complexType name="Fault">
502     <xs:sequence>
503         <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
↳maxOccurs="1" />
504         <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
↳maxOccurs="1" />
505         <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0
↳" maxOccurs="1" />
506         <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↳"1" maxOccurs="1" />
507         <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
↳minOccurs="0" maxOccurs="1" />
508         <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
↳minOccurs="0" />
509     </xs:sequence>
510 </xs:complexType>
511 <xs:simpleType name="FaultSeverityEnum">

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512     <xs:restriction base="xs:string">
513         <xs:enumeration value="low" />
514         <xs:enumeration value="medium" />
515         <xs:enumeration value="high" />
516         <xs:enumeration value="unknown" />
517     </xs:restriction>
518 </xs:simpleType>
519 <xs:simpleType name="Float">
520     <xs:restriction base="xs:float" />
521 </xs:simpleType>
522 <xs:complexType name="GeneralInstructionOrMessageToRoadUsers">
523     <xs:complexContent>
524         <xs:extension base="D2LogicalModel:NetworkManagement">
525             <xs:sequence>
526                 <xs:element name="generalInstructionToRoadUsersType" type=
527                 ↪ "D2LogicalModel:GeneralInstructionToRoadUsersTypeEnum" minOccurs="0" maxOccurs="1" /
528                 ↪ >
529                 <xs:element name="generalMessageToRoadUsers" type=
530                 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
531                 <xs:element name="generalInstructionOrMessageToRoadUsersExtension" type=
532                 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
533             </xs:sequence>
534         </xs:extension>
535     </xs:complexContent>
536 </xs:complexType>
537 <xs:simpleType name="GeneralInstructionToRoadUsersTypeEnum">
538     <xs:restriction base="xs:string">
539         <xs:enumeration value="allowEmergencyVehiclesToPass" />
540         <xs:enumeration value="approachWithCare" />
541         <xs:enumeration value="avoidTheArea" />
542         <xs:enumeration value="closeAllWindowsTurnOffHeaterAndVents" />
543         <xs:enumeration value="crossJunctionWithCare" />
544         <xs:enumeration value="doNotAllowUnnecessaryGaps" />
545         <xs:enumeration value="doNotLeaveYourVehicle" />
546         <xs:enumeration value="doNotThrowOutAnyBurningObjects" />
547         <xs:enumeration value="doNotUseNavigationSystems" />
548         <xs:enumeration value="driveCarefully" />
549         <xs:enumeration value="driveWithExtremeCaution" />
550         <xs:enumeration value="flashYourLights" />
551         <xs:enumeration value="followTheVehicleInFrontSmoothly" />
552         <xs:enumeration value="increaseNormalFollowingDistance" />
553         <xs:enumeration value="inEmergencyWaitForPatrolService" />
554         <xs:enumeration value="keepYourDistance" />
555         <xs:enumeration value="leaveYourVehicleProceedToNextSafePlace" />
556         <xs:enumeration value="noNakedFlames" />
557         <xs:enumeration value="noOvertaking" />
558         <xs:enumeration value="noSmoking" />
559         <xs:enumeration value="noStopping" />
560         <xs:enumeration value="noUturns" />
561         <xs:enumeration value="observeAmberAlert" />
562         <xs:enumeration value="observeSignals" />
563         <xs:enumeration value="observeSigns" />
564         <xs:enumeration value="onlyTravelIfAbsolutelyNecessary" />
565         <xs:enumeration value="overtakeWithCare" />
566         <xs:enumeration value="pullOverToTheEdgeOfTheRoadway" />
567         <xs:enumeration value="stopAtNextSafePlace" />

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564     <xs:enumeration value="stopAtNextServiceArea" />
565     <xs:enumeration value="switchOffEngine" />
566     <xs:enumeration value="switchOffMobilePhonesAndTwoWayRadios" />
567     <xs:enumeration value="testYourBrakes" />
568     <xs:enumeration value="useBusService" />
569     <xs:enumeration value="useFogLights" />
570     <xs:enumeration value="useHazardWarningLights" />
571     <xs:enumeration value="useHeadlights" />
572     <xs:enumeration value="useRailService" />
573     <xs:enumeration value="useTramService" />
574     <xs:enumeration value="useUndergroundService" />
575     <xs:enumeration value="waitForEscortVehicle" />
576     <xs:enumeration value="other" />
577   </xs:restriction>
578 </xs:simpleType>
579 <xs:complexType name="GeneralNetworkManagement">
580   <xs:complexContent>
581     <xs:extension base="D2LogicalModel:NetworkManagement">
582       <xs:sequence>
583         <xs:element name="generalNetworkManagementType" type=
584 ↪ "D2LogicalModel:GeneralNetworkManagementTypeEnum" minOccurs="1" maxOccurs="1" />
585         <xs:element name="trafficManuallyDirectedBy" type=
586 ↪ "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
587         <xs:element name="generalNetworkManagementExtension" type="D2LogicalModel:_
588 ↪ ExtensionType" minOccurs="0" />
589       </xs:sequence>
590     </xs:extension>
591   </xs:complexContent>
592 </xs:complexType>
593 <xs:simpleType name="GeneralNetworkManagementTypeEnum">
594   <xs:restriction base="xs:string">
595     <xs:enumeration value="bridgeSwingInOperation" />
596     <xs:enumeration value="convoyService" />
597     <xs:enumeration value="obstacleSignalling" />
598     <xs:enumeration value="rampMeteringInOperation" />
599     <xs:enumeration value="temporaryTrafficLights" />
600     <xs:enumeration value="tollGatesOpen" />
601     <xs:enumeration value="trafficBeingManuallyDirected" />
602     <xs:enumeration value="trafficHeld" />
603     <xs:enumeration value="other" />
604   </xs:restriction>
605 </xs:simpleType>
606 <xs:simpleType name="GeometryTypeEnum">
607   <xs:restriction base="xs:string" />
608 </xs:simpleType>
609 <xs:complexType name="GlobalNetworkLinear">
610   <xs:sequence>
611     <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1"
612 ↪ maxOccurs="1" />
613     <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"
614 ↪ minOccurs="1" maxOccurs="1" />
615     <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
616 ↪ minOccurs="0" />
617     <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
618 ↪ minOccurs="0" />
619     <xs:element name="linearWithinLinearGNElement" type=

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613   </xs:sequence>
614 </xs:complexType>
615 <xs:complexType name="GrossWeightCharacteristic">
616   <xs:sequence>
617     <xs:element name="comparisonOperator" type=
618       <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
619       <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
620       </xs:sequence>
621 </xs:complexType>
622 <xs:complexType name="GroupOfLocations" abstract="true">
623   <xs:sequence>
624     <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
625     </xs:sequence>
626 </xs:complexType>
627 <xs:complexType name="HeaderInformation">
628   <xs:sequence>
629     <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum"
630     <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum"
631     <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
632     <xs:element name="headerInformationExtension" type="D2LogicalModel:_
633     </xs:sequence>
634 </xs:complexType>
635 <xs:complexType name="HeaviestAxleWeightCharacteristic">
636   <xs:sequence>
637     <xs:element name="comparisonOperator" type=
638     <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
639     <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
640     </xs:sequence>
641 </xs:complexType>
642 <xs:complexType name="HeightCharacteristic">
643   <xs:sequence>
644     <xs:element name="comparisonOperator" type=
645     <xs:element name="vehicleHeight" type="D2LogicalModel:MetresAsFloat" minOccurs=
646     <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
647     </xs:sequence>
648 </xs:complexType>
649 <xs:simpleType name="HeightGradeEnum">
650   <xs:restriction base="xs:string">
651     <xs:enumeration value="aboveGrade" />
652     <xs:enumeration value="atGrade" />
653     <xs:enumeration value="belowGrade" />

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654     </xs:restriction>
655 </xs:simpleType>
656 <xs:complexType name="Impact">
657     <xs:sequence>
658         <xs:element name="capacityRemaining" type="D2LogicalModel:Percentage" minOccurs=
659 ↪ "0" maxOccurs="1" />
660         <xs:element name="numberOfLanesRestricted" type=
661 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
662         <xs:element name="numberOfOperationalLanes" type=
663 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
664         <xs:element name="originalNumberOfLanes" type="D2LogicalModel:NonNegativeInteger
665 ↪ " minOccurs="0" maxOccurs="1" />
666         <xs:element name="trafficConstrictionType" type=
667 ↪ "D2LogicalModel:TrafficConstrictionTypeEnum" minOccurs="0" maxOccurs="1" />
668         <xs:element name="delays" type="D2LogicalModel:Delays" minOccurs="0" />
669         <xs:element name="impactExtension" type="D2LogicalModel:_ExtensionType"
670 ↪ minOccurs="0" />
671     </xs:sequence>
672 </xs:complexType>
673 <xs:simpleType name="InformationStatusEnum">
674     <xs:restriction base="xs:string">
675         <xs:enumeration value="real" />
676         <xs:enumeration value="securityExercise" />
677         <xs:enumeration value="technicalExercise" />
678         <xs:enumeration value="test" />
679     </xs:restriction>
680 </xs:simpleType>
681 <xs:simpleType name="Integer">
682     <xs:restriction base="xs:integer" />
683 </xs:simpleType>
684 <xs:complexType name="InternationalIdentifier">
685     <xs:sequence>
686         <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
687 ↪ maxOccurs="1" />
688         <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1
689 ↪ " maxOccurs="1" />
690         <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
691 ↪ ExtensionType" minOccurs="0" />
692     </xs:sequence>
693 </xs:complexType>
694 <xs:simpleType name="KilometresPerHour">
695     <xs:restriction base="D2LogicalModel:Float" />
696 </xs:simpleType>
697 <xs:simpleType name="LaneEnum">
698     <xs:restriction base="xs:string">
699         <xs:enumeration value="allLanesCompleteCarriageway" />
700         <xs:enumeration value="busLane" />
701         <xs:enumeration value="busStop" />
702         <xs:enumeration value="carPoolLane" />
703         <xs:enumeration value="centralReservation" />
704         <xs:enumeration value="crawlerLane" />
705         <xs:enumeration value="emergencyLane" />
706         <xs:enumeration value="escapeLane" />
707         <xs:enumeration value="expressLane" />
708         <xs:enumeration value="hardShoulder" />
709         <xs:enumeration value="heavyVehicleLane" />

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701     <xs:enumeration value="lane1" />
702     <xs:enumeration value="lane2" />
703     <xs:enumeration value="lane3" />
704     <xs:enumeration value="lane4" />
705     <xs:enumeration value="lane5" />
706     <xs:enumeration value="lane6" />
707     <xs:enumeration value="lane7" />
708     <xs:enumeration value="lane8" />
709     <xs:enumeration value="lane9" />
710     <xs:enumeration value="layBy" />
711     <xs:enumeration value="leftHandTurningLane" />
712     <xs:enumeration value="leftLane" />
713     <xs:enumeration value="localTrafficLane" />
714     <xs:enumeration value="middleLane" />
715     <xs:enumeration value="opposingLanes" />
716     <xs:enumeration value="overtakingLane" />
717     <xs:enumeration value="rightHandTurningLane" />
718     <xs:enumeration value="rightLane" />
719     <xs:enumeration value="rushHourLane" />
720     <xs:enumeration value="setDownArea" />
721     <xs:enumeration value="slowVehicleLane" />
722     <xs:enumeration value="throughTrafficLane" />
723     <xs:enumeration value="tidalFlowLane" />
724     <xs:enumeration value="turningLane" />
725     <xs:enumeration value="verge" />
726 </xs:restriction>
727 </xs:simpleType>
728 <xs:simpleType name="Language">
729     <xs:restriction base="xs:language" />
730 </xs:simpleType>
731 <xs:complexType name="LengthCharacteristic">
732     <xs:sequence>
733         <xs:element name="comparisonOperator" type=
734         ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
735         <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
736         ↪ "1" maxOccurs="1" />
737         <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
738         ↪ ExtensionType" minOccurs="0" />
739     </xs:sequence>
740 </xs:complexType>
741 <xs:complexType name="Linear">
742     <xs:complexContent>
743         <xs:extension base="D2LogicalModel:NetworkLocation">
744             <xs:sequence>
745                 <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
746                 ↪ minOccurs="0" />
747                 <xs:element name="linearWithinLinearElement" type=
748                 ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
749                 <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType
750                 ↪ " minOccurs="0" />
751             </xs:sequence>
752         </xs:extension>
753     </xs:complexContent>
754 </xs:complexType>
755 <xs:complexType name="LinearElement">
756     <xs:sequence>

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751   <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
↪ "0" maxOccurs="1" />
752   <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
753   <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
↪ minOccurs="0" maxOccurs="1" />
754   <xs:element name="linearElementReferenceModelVersion" type=
↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
755   <xs:element name="linearElementNature" type=
↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
756   <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
757   </xs:sequence>
758 </xs:complexType>
759 <xs:complexType name="LinearElementByCode">
760   <xs:complexContent>
761     <xs:extension base="D2LogicalModel:LinearElement">
762       <xs:sequence>
763         <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
764         <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
765       </xs:sequence>
766     </xs:extension>
767   </xs:complexContent>
768 </xs:complexType>
769 <xs:complexType name="LinearElementByPoints">
770   <xs:complexContent>
771     <xs:extension base="D2LogicalModel:LinearElement">
772       <xs:sequence>
773         <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent"
↪ />
774         <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
↪ IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
775         <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
776         <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
777       </xs:sequence>
778     </xs:extension>
779   </xs:complexContent>
780 </xs:complexType>
781 <xs:simpleType name="LinearElementNatureEnum">
782   <xs:restriction base="xs:string">
783     <xs:enumeration value="road" />
784     <xs:enumeration value="roadSection" />
785     <xs:enumeration value="slipRoad" />
786     <xs:enumeration value="other" />
787   </xs:restriction>
788 </xs:simpleType>
789 <xs:simpleType name="LinearReferencingDirectionEnum">
790   <xs:restriction base="xs:string">
791     <xs:enumeration value="both" />
792     <xs:enumeration value="opposite" />
793     <xs:enumeration value="aligned" />
794     <xs:enumeration value="unknown" />
795   </xs:restriction>

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796 </xs:simpleType>
797 <xs:complexType name="LinearWithinLinearElement">
798   <xs:sequence>
799     <xs:element name="administrativeAreaOfLinearSection" type=
800     ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
801     <xs:element name="directionBoundOnLinearSection" type=
802     ↪ "D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
803     <xs:element name="directionRelativeOnLinearSection" type=
804     ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
805     <xs:element name="heightGradeOfLinearSection" type=
806     ↪ "D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
807     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
808     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
809     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
810     <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
811     ↪ ExtensionType" minOccurs="0" />
812   </xs:sequence>
813 </xs:complexType>
814 <xs:complexType name="LinearWithinLinearGNElement">
815   <xs:sequence>
816     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
817     ↪ "0" maxOccurs="1" />
818     <xs:element name="directionRelativeOnLinearSection" type=
819     ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
820     <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
821     ↪ minOccurs="0" maxOccurs="1" />
822     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
823     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
824     <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
825     ↪ ExtensionType" minOccurs="0" />
826   </xs:sequence>
827 </xs:complexType>
828 <xs:simpleType name="LoadTypeEnum">
829   <xs:restriction base="xs:string">
830     <xs:enumeration value="abnormalLoad" />
831     <xs:enumeration value="ammunition" />
832     <xs:enumeration value="chemicals" />
833     <xs:enumeration value="combustibleMaterials" />
834     <xs:enumeration value="corrosiveMaterials" />
835     <xs:enumeration value="debris" />
836     <xs:enumeration value="empty" />
837     <xs:enumeration value="explosiveMaterials" />
838     <xs:enumeration value="extraHighLoad" />
839     <xs:enumeration value="extraLongLoad" />
840     <xs:enumeration value="extraWideLoad" />
841     <xs:enumeration value="fuel" />
842     <xs:enumeration value="glass" />
843     <xs:enumeration value="goods" />
844     <xs:enumeration value="hazardousMaterials" />
845     <xs:enumeration value="liquid" />
846     <xs:enumeration value="livestock" />
847     <xs:enumeration value="materials" />
848     <xs:enumeration value="materialsDangerousForPeople" />
849     <xs:enumeration value="materialsDangerousForTheEnvironment" />
850     <xs:enumeration value="materialsDangerousForWater" />
851     <xs:enumeration value="oil" />

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843     <xs:enumeration value="ordinary" />
844     <xs:enumeration value="perishableProducts" />
845     <xs:enumeration value="petrol" />
846     <xs:enumeration value="pharmaceuticalMaterials" />
847     <xs:enumeration value="radioactiveMaterials" />
848     <xs:enumeration value="refuse" />
849     <xs:enumeration value="toxicMaterials" />
850     <xs:enumeration value="vehicles" />
851     <xs:enumeration value="other" />
852   </xs:restriction>
853 </xs:simpleType>
854 <xs:complexType name="Location" abstract="true">
855   <xs:complexContent>
856     <xs:extension base="D2LogicalModel:GroupOfLocations">
857       <xs:sequence>
858         <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
859           ↪ minOccurs="0" />
860       </xs:sequence>
861     </xs:extension>
862   </xs:complexContent>
863 </xs:complexType>
864 <xs:simpleType name="LocationDescriptorEnum">
865   <xs:restriction base="xs:string">
866     <xs:enumeration value="aroundABendInRoad" />
867     <xs:enumeration value="atMotorwayInterchange" />
868     <xs:enumeration value="atRestArea" />
869     <xs:enumeration value="atServiceArea" />
870     <xs:enumeration value="atTollPlaza" />
871     <xs:enumeration value="atTunnelEntryOrExit" />
872     <xs:enumeration value="inbound" />
873     <xs:enumeration value="inGallery" />
874     <xs:enumeration value="inTheCentre" />
875     <xs:enumeration value="inTheOppositeDirection" />
876     <xs:enumeration value="inTunnel" />
877     <xs:enumeration value="onBorder" />
878     <xs:enumeration value="onBridge" />
879     <xs:enumeration value="onConnector" />
880     <xs:enumeration value="onElevatedSection" />
881     <xs:enumeration value="onFlyover" />
882     <xs:enumeration value="onIceRoad" />
883     <xs:enumeration value="onLevelCrossing" />
884     <xs:enumeration value="onLinkRoad" />
885     <xs:enumeration value="onPass" />
886     <xs:enumeration value="onRoundabout" />
887     <xs:enumeration value="onTheLeft" />
888     <xs:enumeration value="onTheRight" />
889     <xs:enumeration value="onTheRoadway" />
890     <xs:enumeration value="onUndergroundSection" />
891     <xs:enumeration value="onUnderpass" />
892     <xs:enumeration value="outbound" />
893     <xs:enumeration value="overCrestOfHill" />
894     <xs:enumeration value="withinJunction" />
895   </xs:restriction>
896 </xs:simpleType>
897 <xs:simpleType name="MaintenanceVehicleActionsEnum">
  <xs:restriction base="xs:string">

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898     <xs:enumeration value="maintenanceVehiclesMergingIntoTrafficFlow" />
899     <xs:enumeration value="saltAndGritSpreading" />
900     <xs:enumeration value="slowMoving" />
901     <xs:enumeration value="snowClearing" />
902     <xs:enumeration value="stoppingToServiceEquipments" />
903   </xs:restriction>
904 </xs:simpleType>
905 <xs:complexType name="MaintenanceVehicles">
906   <xs:sequence>
907     <xs:element name="numberOfMaintenanceVehicles" type=
908     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
909     <xs:element name="maintenanceVehicleActions" type=
910     ↪ "D2LogicalModel:MaintenanceVehicleActionsEnum" minOccurs="0" maxOccurs="unbounded" /
911     ↪ >
912     <xs:element name="maintenanceVehiclesExtension" type="D2LogicalModel:_
913     ↪ ExtensionType" minOccurs="0" />
914   </xs:sequence>
915 </xs:complexType>
916 <xs:complexType name="MaintenanceWorks">
917   <xs:complexContent>
918     <xs:extension base="D2LogicalModel:Roadworks">
919       <xs:sequence>
920         <xs:element name="roadMaintenanceType" type=
921         ↪ "D2LogicalModel:RoadMaintenanceTypeEnum" minOccurs="1" maxOccurs="unbounded" />
922         <xs:element name="maintenanceWorksExtension" type="D2LogicalModel:_
923         ↪ ExtensionType" minOccurs="0" />
924       </xs:sequence>
925     </xs:extension>
926   </xs:complexContent>
927 </xs:complexType>
928 <xs:complexType name="MeasurementEquipmentFault">
929   <xs:complexContent>
930     <xs:extension base="D2LogicalModel:Fault">
931       <xs:sequence>
932         <xs:element name="measurementEquipmentFault" type=
933         ↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
934         <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
935         ↪ ExtensionType" minOccurs="0" />
936       </xs:sequence>
937     </xs:extension>
938   </xs:complexContent>
939 </xs:complexType>
940 <xs:simpleType name="MeasurementEquipmentFaultEnum">
941   <xs:restriction base="xs:string">
942     <xs:enumeration value="intermittentDataValues" />
943     <xs:enumeration value="noDataValuesAvailable" />
944     <xs:enumeration value="spuriousUnreliableDataValues" />
945     <xs:enumeration value="unspecifiedOrUnknownFault" />
946     <xs:enumeration value="other" />
947   </xs:restriction>
948 </xs:simpleType>
949 <xs:simpleType name="MetresAsFloat">
950   <xs:restriction base="D2LogicalModel:Float" />
951 </xs:simpleType>
952 <xs:simpleType name="MetresAsNonNegativeInteger">
953   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />

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946 </xs:simpleType>
947 <xs:complexType name="Mobility">
948   <xs:sequence>
949     <xs:element name="mobilityType" type="D2LogicalModel:MobilityEnum" minOccurs="1
↪ " maxOccurs="1" />
950     <xs:element name="mobilityExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
951   </xs:sequence>
952 </xs:complexType>
953 <xs:simpleType name="MobilityEnum">
954   <xs:restriction base="xs:string">
955     <xs:enumeration value="mobile" />
956     <xs:enumeration value="stationary" />
957     <xs:enumeration value="unknown" />
958   </xs:restriction>
959 </xs:simpleType>
960 <xs:complexType name="MultilingualString">
961   <xs:sequence>
962     <xs:element name="values">
963       <xs:complexType>
964         <xs:sequence>
965           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
↪ maxOccurs="unbounded" />
966         </xs:sequence>
967       </xs:complexType>
968     </xs:element>
969   </xs:sequence>
970 </xs:complexType>
971 <xs:complexType name="MultilingualStringValue">
972   <xs:simpleContent>
973     <xs:extension base="D2LogicalModel:MultilingualStringValueType">
974       <xs:attribute name="lang" type="xs:language" />
975     </xs:extension>
976   </xs:simpleContent>
977 </xs:complexType>
978 <xs:simpleType name="MultilingualStringValueType">
979   <xs:restriction base="xs:string">
980     <xs:maxLength value="1024" />
981   </xs:restriction>
982 </xs:simpleType>
983 <xs:complexType name="NetworkLocation" abstract="true">
984   <xs:complexContent>
985     <xs:extension base="D2LogicalModel:Location">
986       <xs:sequence>
987         <xs:element name="supplementaryPositionalDescription" type=
↪ "D2LogicalModel:SupplementaryPositionalDescription" minOccurs="0" />
988         <xs:element name="destination" type="D2LogicalModel:Destination" minOccurs=
↪ "0" />
989         <xs:element name="networkLocationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
990       </xs:sequence>
991     </xs:extension>
992   </xs:complexContent>
993 </xs:complexType>
994 <xs:complexType name="NetworkManagement" abstract="true">
995   <xs:complexContent>

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996     <xs:extension base="D2LogicalModel:OperatorAction">
997         <xs:sequence>
998             <xs:element name="complianceOption" type=
999 ↪ "D2LogicalModel:ComplianceOptionEnum" minOccurs="1" maxOccurs="1" />
1000             <xs:element name="applicableForTrafficDirection" type=
1001 ↪ "D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="unbounded" />
1002             <xs:element name="applicableForTrafficType" type=
1003 ↪ "D2LogicalModel:TrafficTypeEnum" minOccurs="0" maxOccurs="unbounded" />

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1001     <xs:element name="placesAtWhichApplicable" type="D2LogicalModel:PlacesEnum" ↪
1002 ↪ minOccurs="0" maxOccurs="unbounded" />
1003     <xs:element name="forVehiclesWithCharacteristicsOf" type=
1004 ↪ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" maxOccurs="unbounded" />
1005     <xs:element name="networkManagementExtension" type="D2LogicalModel:_
1006 ↪ ExtensionType" minOccurs="0" />
1007     </xs:sequence>
1008     </xs:extension>
1009     </xs:complexContent>
1010     </xs:complexType>
1011     <xs:complexType name="NonManagedCause">
1012     <xs:complexContent>
1013     <xs:extension base="D2LogicalModel:Cause">
1014     <xs:sequence>
1015     <xs:element name="causeType" type="D2LogicalModel:CauseTypeEnum" minOccurs=
1016 ↪ "0" maxOccurs="1" />
1017     <xs:element name="nonManagedCauseExtension" type="D2LogicalModel:_
1018 ↪ ExtensionType" minOccurs="0" />
1019     </xs:sequence>
1020     </xs:extension>
1021     </xs:complexContent>
1022     </xs:complexType>
1023     <xs:simpleType name="NonNegativeInteger">
1024     <xs:restriction base="xs:nonNegativeInteger" />
1025     </xs:simpleType>
1026     <xs:complexType name="NumberOfAxlesCharacteristic">
1027     <xs:sequence>
1028     <xs:element name="comparisonOperator" type=
1029 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1030     <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger" ↪
1031 ↪ minOccurs="1" maxOccurs="1" />
1032     <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
1033 ↪ ExtensionType" minOccurs="0" />
1034     </xs:sequence>
1035     </xs:complexType>
1036     <xs:complexType name="OccupancyChangeValue">
1037     <xs:complexContent>
1038     <xs:extension base="D2LogicalModel:DataValue">
1039     <xs:sequence>
1040     <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
1041 ↪ "1" maxOccurs="1" />
1042     <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
1043 ↪ ExtensionType" minOccurs="0" />
1044     </xs:sequence>
1045     </xs:extension>
1046     </xs:complexContent>
1047     </xs:complexType>

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1038 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
1039   <xs:sequence>
1040     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1041     <xs:element name="openlrLineAttributes" type=
1042     ↪ "D2LogicalModel:OpenlrLineAttributes" />
1043     <xs:element name="openlrBaseLocationReferencePointExtension" type=
1044     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1045   </xs:sequence>
1046 </xs:complexType>
1047 <xs:complexType name="OpenlrBasePointLocation" abstract="true">
1048   <xs:sequence>
1049     <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
1050     ↪ minOccurs="1" maxOccurs="1" />
1051     <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
1052     ↪ minOccurs="1" maxOccurs="1" />
1053     <xs:element name="openlrPositiveOffset" type=
1054     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1055     <xs:element name="openlrLocationReferencePoint" type=
1056     ↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
1057     <xs:element name="openlrLastLocationReferencePoint" type=
1058     ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1059     <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
1060     ↪ ExtensionType" minOccurs="0" />
1061   </xs:sequence>
1062 </xs:complexType>
1063 <xs:complexType name="OpenlrExtendedLinear">
1064   <xs:sequence>
1065     <xs:element name="firstDirection" type=
1066     ↪ "D2LogicalModel:OpenlrLineLocationReference" />
1067     <xs:element name="oppositeDirection" type=
1068     ↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
1069   </xs:sequence>
1070 </xs:complexType>
1071 <xs:complexType name="OpenlrExtendedPoint">
1072   <xs:sequence>
1073     <xs:element name="openlrPointLocationReference" type=
1074     ↪ "D2LogicalModel:OpenlrPointLocationReference" />
1075   </xs:sequence>
1076 </xs:complexType>
1077 <xs:simpleType name="OpenlrFormOfWayEnum">
1078   <xs:restriction base="xs:string">
1079     <xs:enumeration value="undefined" />
1080     <xs:enumeration value="motorway" />
1081     <xs:enumeration value="multipleCarriageway" />
1082     <xs:enumeration value="singleCarriageway" />
1083     <xs:enumeration value="roundabout" />
1084     <xs:enumeration value="slipRoad" />
1085     <xs:enumeration value="trafficSquare" />
1086     <xs:enumeration value="other" />
1087   </xs:restriction>
1088 </xs:simpleType>
1089 <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
1090   <xs:restriction base="xs:string">
1091     <xs:enumeration value="FRC0" />
1092     <xs:enumeration value="FRC1" />
1093     <xs:enumeration value="FRC2" />

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1083     <xs:enumeration value="FRC3" />
1084     <xs:enumeration value="FRC4" />
1085     <xs:enumeration value="FRC5" />
1086     <xs:enumeration value="FRC6" />
1087     <xs:enumeration value="FRC7" />
1088   </xs:restriction>
1089 </xs:simpleType>
1090 <xs:complexType name="OpenlrGeoCoordinate">
1091   <xs:sequence>
1092     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1093     <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
1094 ↪ ExtensionType" minOccurs="0" />
1095   </xs:sequence>
1096 </xs:complexType>
1097 <xs:complexType name="OpenlrLastLocationReferencePoint">
1098   <xs:complexContent>
1099     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1100       <xs:sequence>
1101         <xs:element name="openlrLastLocationReferencePointExtension" type=
1102 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1103       </xs:sequence>
1104     </xs:extension>
1105   </xs:complexContent>
1106 </xs:complexType>
1107 <xs:complexType name="OpenlrLineAttributes">
1108   <xs:sequence>
1109     <xs:element name="openlrFunctionalRoadClass" type=
1110 ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1111     <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
1112 ↪ minOccurs="1" maxOccurs="1" />
1113     <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
1114 ↪ "1" maxOccurs="1" />
1115     <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
1116 ↪ ExtensionType" minOccurs="0" />
1117   </xs:sequence>
1118 </xs:complexType>
1119 <xs:complexType name="OpenlrLineLocationReference">
1120   <xs:sequence>
1121     <xs:element name="openlrLocationReferencePoint" type=
1122 ↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
1123     <xs:element name="openlrLastLocationReferencePoint" type=
1124 ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1125     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
1126 ↪ "0" />
1127     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
1128 ↪ ExtensionType" minOccurs="0" />
1129   </xs:sequence>
1130 </xs:complexType>
1131 <xs:complexType name="OpenlrLocationReferencePoint">
1132   <xs:complexContent>
1133     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1134       <xs:sequence>
1135         <xs:element name="openlrPathAttributes" type=
1136 ↪ "D2LogicalModel:OpenlrPathAttributes" />
1137         <xs:element name="openlrLocationReferencePointExtension" type=
1138 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />

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1127     </xs:sequence>
1128   </xs:extension>
1129 </xs:complexContent>
1130 </xs:complexType>
1131 <xs:complexType name="OpenlrOffsets">
1132   <xs:sequence>
1133     <xs:element name="openlrPositiveOffset" type=
1134 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1135     <xs:element name="openlrNegativeOffset" type=
1136 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1137     <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_ExtensionType"
1138 ↪ minOccurs="0" />
1139   </xs:sequence>
1140 </xs:complexType>
1141 <xs:simpleType name="OpenlrOrientationEnum">
1142   <xs:restriction base="xs:string">
1143     <xs:enumeration value="noOrientationOrUnknown" />
1144     <xs:enumeration value="withLineDirection" />
1145     <xs:enumeration value="againstLineDirection" />
1146     <xs:enumeration value="both" />
1147   </xs:restriction>
1148 </xs:simpleType>
1149 <xs:complexType name="OpenlrPathAttributes">
1150   <xs:sequence>
1151     <xs:element name="openlrLowestFRCToNextLRPoint" type=
1152 ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1153     <xs:element name="openlrDistanceToNextLRPoint" type=
1154 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
1155     <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
1156 ↪ ExtensionType" minOccurs="0" />
1157   </xs:sequence>
1158 </xs:complexType>
1159 <xs:complexType name="OpenlrPointAlongLine">
1160   <xs:complexContent>
1161     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1162       <xs:sequence>
1163         <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
1164 ↪ ExtensionType" minOccurs="0" />
1165       </xs:sequence>
1166     </xs:extension>
1167   </xs:complexContent>
1168 </xs:complexType>
1169 <xs:complexType name="OpenlrPointLocationReference">
1170   <xs:sequence>
1171     <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
1172 ↪ " minOccurs="0" />
1173     <xs:element name="openlrPoiWithAccessPoint" type=
1174 ↪ "D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
1175     <xs:element name="openlrPointAlongLine" type=
1176 ↪ "D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
1177     <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
1178 ↪ ExtensionType" minOccurs="0" />
1179   </xs:sequence>
1180 </xs:complexType>
1181 <xs:complexType name="OpenlrPoiWithAccessPoint">
1182   <xs:complexContent>

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1172     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1173         <xs:sequence>
1174             <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
1175             <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
1176             <ExtensionType" minOccurs="0" />
1177         </xs:sequence>
1178     </xs:extension>
1179 </xs:complexContent>
1180 </xs:complexType>
1181 <xs:simpleType name="OpenlrSideOfRoadEnum">
1182     <xs:restriction base="xs:string">
1183         <xs:enumeration value="onRoadOrUnknown" />
1184         <xs:enumeration value="right" />
1185         <xs:enumeration value="left" />
1186         <xs:enumeration value="both" />
1187     </xs:restriction>
1188 </xs:simpleType>
1189 <xs:complexType name="OperatorAction">
1190     <xs:complexContent>
1191         <xs:extension base="D2LogicalModel:SituationRecord">
1192             <xs:sequence>
1193                 <xs:element name="operatorActionExtension" type="D2LogicalModel:_
1194                 <ExtensionType" minOccurs="0" />
1195             </xs:sequence>
1196         </xs:extension>
1197     </xs:complexContent>
1198 </xs:complexType>
1199 <xs:complexType name="OverallPeriod">
1200     <xs:sequence>
1201         <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs="1
1202         <maxOccurs="1" />
1203         <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0"
1204         <maxOccurs="1" />
1205         <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType"
1206         <minOccurs="0" />
1207     </xs:sequence>
1208 </xs:complexType>
1209 <xs:simpleType name="PassengerCarUnitsPerHour">
1210     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1211 </xs:simpleType>
1212 <xs:complexType name="PayloadPublication" abstract="true">
1213     <xs:sequence>
1214         <xs:element name="feedDescription" type="D2LogicalModel:MultilingualString"
1215         <minOccurs="0" maxOccurs="1" />
1216         <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
1217         <maxOccurs="1" />
1218         <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
1219         <maxOccurs="1" />
1220         <xs:element name="publicationCreator" type=
1221         <"D2LogicalModel:InternationalIdentifier" />
1222         <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
1223         <ExtensionType" minOccurs="0" />
1224     </xs:sequence>
1225     <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
1226 </xs:complexType>

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1217 <xs:complexType name="PcuFlowValue">
1218   <xs:complexContent>
1219     <xs:extension base="D2LogicalModel:DataValue">
1220       <xs:sequence>
1221         <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
1222 ↪ " minOccurs="1" maxOccurs="1" />
1223         <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
1224 ↪ " minOccurs="0" />
1225       </xs:sequence>
1226     </xs:extension>
1227   </xs:complexContent>
1228 </xs:complexType>
1229 <xs:simpleType name="Percentage">
1230   <xs:restriction base="D2LogicalModel:Float" />
1231 </xs:simpleType>
1232 <xs:complexType name="PercentageDistanceAlongLinearElement">
1233   <xs:complexContent>
1234     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
1235       <xs:sequence>
1236         <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage"
1237 ↪ minOccurs="1" maxOccurs="1" />
1238         <xs:element name="percentageDistanceAlongLinearElementExtension" type=
1239 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1240       </xs:sequence>
1241     </xs:extension>
1242   </xs:complexContent>
1243 </xs:complexType>
1244 <xs:simpleType name="PersonCategoryEnum">
1245   <xs:restriction base="xs:string">
1246     <xs:enumeration value="adult" />
1247     <xs:enumeration value="child" />
1248     <xs:enumeration value="emergencyServicesPerson" />
1249     <xs:enumeration value="fireman" />
1250     <xs:enumeration value="infant" />
1251     <xs:enumeration value="medicalStaff" />
1252     <xs:enumeration value="memberOfThePublic" />
1253     <xs:enumeration value="policeman" />
1254     <xs:enumeration value="politician" />
1255     <xs:enumeration value="publicTransportPassenger" />
1256     <xs:enumeration value="sickPerson" />
1257     <xs:enumeration value="trafficOfficer" />
1258     <xs:enumeration value="trafficWarden" />
1259     <xs:enumeration value="veryImportantPerson" />
1260   </xs:restriction>
1261 </xs:simpleType>
1262 <xs:simpleType name="PlacesEnum">
1263   <xs:restriction base="xs:string">
1264     <xs:enumeration value="aroundBendsInTheRoad" />
1265     <xs:enumeration value="atCustomsPosts" />
1266     <xs:enumeration value="atHighAltitudes" />
1267     <xs:enumeration value="atTollPlazas" />
1268     <xs:enumeration value="inBuiltUpAreas" />
1269     <xs:enumeration value="inContraflowSections" />
1270     <xs:enumeration value="inForestedAreas" />
1271     <xs:enumeration value="inGalleries" />
1272     <xs:enumeration value="inLowLyingAreas" />

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1269     <xs:enumeration value="inRoadworksAreas" />
1270     <xs:enumeration value="inRuralAreas" />
1271     <xs:enumeration value="inShadedAreas" />
1272     <xs:enumeration value="inTheCityCentre" />
1273     <xs:enumeration value="inTheInnerCityAreas" />
1274     <xs:enumeration value="inTunnels" />
1275     <xs:enumeration value="onBridges" />
1276     <xs:enumeration value="onDownHillSections" />
1277     <xs:enumeration value="onDualCarriagewaySections" />
1278     <xs:enumeration value="onElevatedSections" />
1279     <xs:enumeration value="onEnteringOrLeavingTunnels" />
1280     <xs:enumeration value="onEnteringTheCountry" />
1281     <xs:enumeration value="onFlyovers" />
1282     <xs:enumeration value="onLeavingTheCountry" />
1283     <xs:enumeration value="onMotorways" />
1284     <xs:enumeration value="onNonMotorways" />
1285     <xs:enumeration value="onPasses" />
1286     <xs:enumeration value="onRoundabouts" />
1287     <xs:enumeration value="onSingleCarriagewaySections" />
1288     <xs:enumeration value="onSlipRoads" />
1289     <xs:enumeration value="onUndergroundSections" />
1290     <xs:enumeration value="onUnderpasses" />
1291     <xs:enumeration value="onUpHillSections" />
1292     <xs:enumeration value="overTheCrestOfHills" />
1293     <xs:enumeration value="other" />
1294 </xs:restriction>
1295 </xs:simpleType>
1296 <xs:complexType name="Point">
1297     <xs:complexContent>
1298         <xs:extension base="D2LogicalModel:NetworkLocation">
1299             <xs:sequence>
1300                 <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
1301 ↪ "0" />
1302                 <xs:element name="pointAlongLinearElement" type=
1303 ↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
1304                 <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
1305 ↪ minOccurs="0" />
1306             </xs:sequence>
1307         </xs:extension>
1308     </xs:complexContent>
1309 </xs:complexType>
1310 <xs:complexType name="PointAlongLinearElement">
1311     <xs:sequence>
1312         <xs:element name="administrativeAreaOfPoint" type=
1313 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1314         <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
1315 ↪ minOccurs="0" maxOccurs="1" />
1316         <xs:element name="directionRelativeAtPoint" type=
1317 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
1318         <xs:element name="heightGradeOfPoint" type="D2LogicalModel:HeightGradeEnum"
1319 ↪ minOccurs="0" maxOccurs="1" />
1320         <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1321         <xs:element name="distanceAlongLinearElement" type=
1322 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1323         <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
1324 ↪ ExtensionType" minOccurs="0" />

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1316     </xs:sequence>
1317 </xs:complexType>
1318 <xs:complexType name="PointByMultiCoordinates">
1319     <xs:sequence>
1320         <xs:element name="sjtskPointCoordinates" type=
1321 ↪ "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
1322         <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" ↪
1323 ↪ minOccurs="0" />
1324     </xs:sequence>
1325 </xs:complexType>
1326 <xs:complexType name="PointCoordinates">
1327     <xs:sequence>
1328         <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
1329 ↪ "1" />
1330         <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1" ↪
1331 ↪ maxOccurs="1" />
1332         <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
1333 ↪ " minOccurs="0" />
1334     </xs:sequence>
1335 </xs:complexType>
1336 <xs:complexType name="PointDestination">
1337     <xs:complexContent>
1338         <xs:extension base="D2LogicalModel:Destination">
1339             <xs:sequence>
1340                 <xs:element name="point" type="D2LogicalModel:Point" />
1341                 <xs:element name="pointDestinationExtension" type="D2LogicalModel:_
1342 ↪ ExtensionType" minOccurs="0" />
1343             </xs:sequence>
1344         </xs:extension>
1345     </xs:complexContent>
1346 </xs:complexType>
1347 <xs:complexType name="PointWithinLinearGNElement">
1348     <xs:sequence>
1349         <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1350 ↪ "0" maxOccurs="1" />
1351         <xs:element name="directionRelativeOnLinearSection" type=
1352 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1353         <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1" ↪
1354 ↪ maxOccurs="1" />
1355         <xs:element name="distanceAlongLinearElement" type=
1356 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1357     </xs:sequence>
1358 </xs:complexType>
1359 <xs:simpleType name="ProbabilityOfOccurrenceEnum">
1360     <xs:restriction base="xs:string">
1361         <xs:enumeration value="certain" />
1362         <xs:enumeration value="probable" />
1363         <xs:enumeration value="riskOf" />
1364     </xs:restriction>
1365 </xs:simpleType>
1366 <xs:complexType name="Referent">
1367     <xs:sequence>
1368         <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1
1369 ↪ " maxOccurs="1" />
1370         <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0" ↪
1371 ↪ maxOccurs="1" />

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1360   <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
↪ minOccurs="1" maxOccurs="1" />
1361   <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
1362   <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
↪ minOccurs="0" />
1363   <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1364   </xs:sequence>
1365 </xs:complexType>
1366 <xs:simpleType name="ReferentTypeEnum">
1367   <xs:restriction base="xs:string">
1368     <xs:enumeration value="boundary" />
1369     <xs:enumeration value="intersection" />
1370     <xs:enumeration value="referenceMarker" />
1371     <xs:enumeration value="landmark" />
1372     <xs:enumeration value="roadNode" />
1373   </xs:restriction>
1374 </xs:simpleType>
1375 <xs:complexType name="ReroutingManagement">
1376   <xs:complexContent>
1377     <xs:extension base="D2LogicalModel:NetworkManagement">
1378       <xs:sequence>
1379         <xs:element name="reroutingManagementType" type=
↪ "D2LogicalModel:ReroutingManagementTypeEnum" minOccurs="1" maxOccurs="unbounded" />
1380         <xs:element name="reroutingItineraryDescription" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1381         <xs:element name="signedRerouting" type="D2LogicalModel:Boolean" minOccurs=
↪ "0" maxOccurs="1" />
1382         <xs:element name="entry" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
1383         <xs:element name="exit" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
1384         <xs:element name="roadOrJunctionNumber" type="D2LogicalModel:String"
↪ minOccurs="0" maxOccurs="1" />
1385         <xs:element name="reroutingManagementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1386       </xs:sequence>
1387     </xs:extension>
1388   </xs:complexContent>
1389 </xs:complexType>
1390 <xs:simpleType name="ReroutingManagementTypeEnum">
1391   <xs:restriction base="xs:string">
1392     <xs:enumeration value="doNotFollowDiversionSigns" />
1393     <xs:enumeration value="doNotUseEntry" />
1394     <xs:enumeration value="doNotUseExit" />
1395     <xs:enumeration value="doNotUseIntersectionOrJunction" />
1396     <xs:enumeration value="followDiversionSigns" />
1397     <xs:enumeration value="followLocalDiversion" />
1398     <xs:enumeration value="followSpecialMarkers" />
1399     <xs:enumeration value="useEntry" />
1400     <xs:enumeration value="useExit" />
1401     <xs:enumeration value="useIntersectionOrJunction" />
1402   </xs:restriction>
1403 </xs:simpleType>
1404 <xs:simpleType name="RoadMaintenanceTypeEnum">

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1405 <xs:restriction base="xs:string">
1406   <xs:enumeration value="clearanceWork" />
1407   <xs:enumeration value="controlledAvalanche" />
1408   <xs:enumeration value="installationWork" />
1409   <xs:enumeration value="grassCuttingWork" />
1410   <xs:enumeration value="litterClearance" />
1411   <xs:enumeration value="maintenanceWork" />
1412   <xs:enumeration value="overheadWorks" />
1413   <xs:enumeration value="repairWork" />
1414   <xs:enumeration value="resurfacingWork" />
1415   <xs:enumeration value="roadMarkingWork" />
1416   <xs:enumeration value="roadsideWork" />
1417   <xs:enumeration value="roadworksClearance" />
1418   <xs:enumeration value="roadworks" />
1419   <xs:enumeration value="rockFallPreventativeMaintenance" />
1420   <xs:enumeration value="saltingInProgress" />
1421   <xs:enumeration value="snowploughsInUse" />
1422   <xs:enumeration value="sweepingOfRoad" />
1423   <xs:enumeration value="treeAndVegetationCuttingWork" />
1424   <xs:enumeration value="other" />
1425 </xs:restriction>
1426 </xs:simpleType>
1427 <xs:complexType name="RoadOrCarriagewayOrLaneManagement">
1428   <xs:complexContent>
1429     <xs:extension base="D2LogicalModel:NetworkManagement">
1430       <xs:sequence>
1431         <xs:element name="roadOrCarriagewayOrLaneManagementType" type=
1432 ↪ "D2LogicalModel:RoadOrCarriagewayOrLaneManagementTypeEnum" minOccurs="1" maxOccurs=
1433 ↪ "1" />
1434         <xs:element name="minimumCarOccupancy" type=
1435 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1436         <xs:element name="roadOrCarriagewayOrLaneManagementExtension" type=
1437 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1438       </xs:sequence>
1439     </xs:extension>
1440   </xs:complexContent>
1441 </xs:complexType>
1442 <xs:simpleType name="RoadOrCarriagewayOrLaneManagementTypeEnum">
1443   <xs:restriction base="xs:string">
1444     <xs:enumeration value="carPoolLaneInOperation" />
1445     <xs:enumeration value="carriagewayClosures" />
1446     <xs:enumeration value="clearALaneForEmergencyVehicles" />
1447     <xs:enumeration value="clearALaneForSnowploughsAndGrittingVehicles" />
1448     <xs:enumeration value="closedPermanentlyForTheWinter" />
1449     <xs:enumeration value="contraflow" />
1450     <xs:enumeration value="doNotUseSpecifiedLanesOrCarriageways" />
1451     <xs:enumeration value="hardShoulderRunningInOperation" />
1452     <xs:enumeration value="heightRestrictionInOperation" />
1453     <xs:enumeration value="intermittentShortTermClosures" />
1454     <xs:enumeration value="keepToTheLeft" />
1455     <xs:enumeration value="keepToTheRight" />
1456     <xs:enumeration value="laneClosures" />
1457     <xs:enumeration value="lanesDeviated" />
1458     <xs:enumeration value="narrowLanes" />
1459     <xs:enumeration value="newRoadworksLayout" />
1460     <xs:enumeration value="overnightClosures" />

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1457     <xs:enumeration value="roadCleared" />
1458     <xs:enumeration value="roadClosed" />
1459     <xs:enumeration value="rollingRoadBlock" />
1460     <xs:enumeration value="rushHourLaneInOperation" />
1461     <xs:enumeration value="singleAlternateLineTraffic" />
1462     <xs:enumeration value="tidalFlowLaneInOperation" />
1463     <xs:enumeration value="turnAroundInOperation" />
1464     <xs:enumeration value="useOfSpecifiedLanesOrCarriagewaysAllowed" />
1465     <xs:enumeration value="useSpecifiedLanesOrCarriageways" />
1466     <xs:enumeration value="vehicleStorageInOperation" />
1467     <xs:enumeration value="weightRestrictionInOperation" />
1468     <xs:enumeration value="other" />
1469   </xs:restriction>
1470 </xs:simpleType>
1471 <xs:complexType name="RoadsideAssistance">
1472   <xs:complexContent>
1473     <xs:extension base="D2LogicalModel:OperatorAction">
1474       <xs:sequence>
1475         <xs:element name="roadsideAssistanceType" type=
1476 ↪ "D2LogicalModel:RoadsideAssistanceTypeEnum" minOccurs="1" maxOccurs="1" />
1477         <xs:element name="roadsideAssistanceExtension" type="D2LogicalModel:_
1478 ↪ ExtensionType" minOccurs="0" />
1479       </xs:sequence>
1480     </xs:extension>
1481   </xs:complexContent>
1482 </xs:complexType>
1483 <xs:simpleType name="RoadsideAssistanceTypeEnum">
1484   <xs:restriction base="xs:string">
1485     <xs:enumeration value="airAmbulance" />
1486     <xs:enumeration value="busPassengerAssistance" />
1487     <xs:enumeration value="emergencyServices" />
1488     <xs:enumeration value="firstAid" />
1489     <xs:enumeration value="foodDelivery" />
1490     <xs:enumeration value="helicopterRescue" />
1491     <xs:enumeration value="vehicleRepair" />
1492     <xs:enumeration value="vehicleRecovery" />
1493     <xs:enumeration value="other" />
1494   </xs:restriction>
1495 </xs:simpleType>
1496 <xs:complexType name="Roadworks" abstract="true">
1497   <xs:complexContent>
1498     <xs:extension base="D2LogicalModel:OperatorAction">
1499       <xs:sequence>
1500 ↪ <xs:element name="roadworksDuration" type=
1501 ↪ "D2LogicalModel:RoadworksDurationEnum" minOccurs="0" maxOccurs="1" />
1502 ↪ <xs:element name="roadworksScale" type="D2LogicalModel:RoadworksScaleEnum"
1503 ↪ minOccurs="0" maxOccurs="1" />
1504 ↪ <xs:element name="underTraffic" type="D2LogicalModel:Boolean" minOccurs="0"
1505 ↪ maxOccurs="1" />
1506 ↪ <xs:element name="urgentRoadworks" type="D2LogicalModel:Boolean" minOccurs=
1507 ↪ "0" maxOccurs="1" />
1508 ↪ <xs:element name="mobility" type="D2LogicalModel:Mobility" minOccurs="0" />
1509 ↪ <xs:element name="subjects" type="D2LogicalModel:Subjects" minOccurs="0" />
1510 ↪ <xs:element name="maintenanceVehicles" type=
1511 ↪ "D2LogicalModel:MaintenanceVehicles" minOccurs="0" />
1512 ↪ <xs:element name="roadworksExtension" type="D2LogicalModel:_ExtensionType"

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1506     </xs:sequence>
1507   </xs:extension>
1508 </xs:complexContent>
1509 </xs:complexType>
1510 <xs:simpleType name="RoadworksDurationEnum">
1511   <xs:restriction base="xs:string">
1512     <xs:enumeration value="longTerm" />
1513     <xs:enumeration value="mediumTerm" />
1514     <xs:enumeration value="shortTerm" />
1515   </xs:restriction>
1516 </xs:simpleType>
1517 <xs:simpleType name="RoadworksScaleEnum">
1518   <xs:restriction base="xs:string">
1519     <xs:enumeration value="major" />
1520     <xs:enumeration value="medium" />
1521     <xs:enumeration value="minor" />
1522   </xs:restriction>
1523 </xs:simpleType>
1524 <xs:simpleType name="Seconds">
1525   <xs:restriction base="D2LogicalModel:Float" />
1526 </xs:simpleType>
1527 <xs:complexType name="Situation">
1528   <xs:sequence>
1529     <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"
1530     <minOccurs="0" maxOccurs="1" />
1531     <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
1532     <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"
1533     <maxOccurs="unbounded" />
1534     <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"
1535     <minOccurs="0" />
1536   </xs:sequence>
1537   <xs:attribute name="id" type="xs:string" use="required" />
1538   <xs:attribute name="version" type="xs:string" use="required" />
1539 </xs:complexType>
1540 <xs:complexType name="SituationPublication">
1541   <xs:complexContent>
1542     <xs:extension base="D2LogicalModel:PayloadPublication">
1543       <xs:sequence>
1544         <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0"
1545         <maxOccurs="unbounded" />
1546         <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
1547         <ExtensionType" minOccurs="0" />
1548       </xs:sequence>
1549     </xs:extension>
1550   </xs:complexContent>
1551 </xs:complexType>
1552 <xs:complexType name="SituationRecord" abstract="true">
1553   <xs:sequence>
1554     <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime"
1555     <minOccurs="1" maxOccurs="1" />
1556     <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"
1557     <minOccurs="1" maxOccurs="1" />
1558     <xs:element name="probabilityOfOccurrence" type=
1559     <"D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1" />
1560     <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />

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1553     <xs:element name="validity" type="D2LogicalModel:Validity" />
1554     <xs:element name="impact" type="D2LogicalModel:Impact" minOccurs="0" />
1555     <xs:element name="cause" type="D2LogicalModel:Cause" minOccurs="0" />
1556     <xs:element name="generalPublicComment" type="D2LogicalModel:Comment" minOccurs=
1557     ↪ "0" maxOccurs="unbounded" />
1558     <xs:element name="urlLink" type="D2LogicalModel:UrlLink" minOccurs="0" ↪
1559     ↪ maxOccurs="unbounded" />
1560     <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
1561     <xs:element name="situationRecordExtension" type="D2LogicalModel:_
1562     ↪ SituationRecordExtensionType" minOccurs="0" />
1563   </xs:sequence>
1564   <xs:attribute name="id" type="xs:string" use="required" />
1565   <xs:attribute name="version" type="xs:string" use="required" />
1566 </xs:complexType>
1567 <xs:complexType name="SituationRecordExtendedApproved">
1568   <xs:sequence>
1569     <xs:element name="safetyRelatedMessage" type="D2LogicalModel:Boolean" minOccurs=
1570     ↪ "0" maxOccurs="1" />
1571   </xs:sequence>
1572 </xs:complexType>
1573 <xs:complexType name="SJTSKPointCoordinates">
1574   <xs:sequence>
1575     <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1576     ↪ " />
1577     <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1578     ↪ " />
1579     <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
1580     ↪ ExtensionType" minOccurs="0" />
1581   </xs:sequence>
1582 </xs:complexType>
1583 <xs:complexType name="Source">
1584   <xs:sequence>
1585     <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs="0"
1586     ↪ " maxOccurs="1" />
1587     <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
1588     ↪ "0" maxOccurs="1" />
1589     <xs:element name="sourceName" type="D2LogicalModel:MultilingualString" ↪
1590     ↪ minOccurs="0" maxOccurs="1" />
1591     <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs="0"
1592     ↪ " maxOccurs="1" />
1593     <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0" ↪
1594     ↪ maxOccurs="1" />
1595     <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType" ↪
1596     ↪ minOccurs="0" />
1597   </xs:sequence>
1598 </xs:complexType>
1599 <xs:simpleType name="SourceTypeEnum">
1600   <xs:restriction base="xs:string">
1601     <xs:enumeration value="automobileClubPatrol" />
1602     <xs:enumeration value="cameraObservation" />
1603     <xs:enumeration value="freightVehicleOperator" />
1604     <xs:enumeration value="inductionLoopMonitoringStation" />
1605     <xs:enumeration value="infraredMonitoringStation" />
1606     <xs:enumeration value="microwaveMonitoringStation" />
1607     <xs:enumeration value="mobileTelephoneCaller" />
1608     <xs:enumeration value="nonPoliceEmergencyServicePatrol" />

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1596     <xs:enumeration value="otherInformation" />
1597     <xs:enumeration value="otherOfficialVehicle" />
1598     <xs:enumeration value="policePatrol" />
1599     <xs:enumeration value="privateBreakdownService" />
1600     <xs:enumeration value="publicAndPrivateUtilities" />
1601     <xs:enumeration value="registeredMotoristObserver" />
1602     <xs:enumeration value="roadAuthorities" />
1603     <xs:enumeration value="roadOperatorPatrol" />
1604     <xs:enumeration value="roadsideTelephoneCaller" />
1605     <xs:enumeration value="spotterAircraft" />
1606     <xs:enumeration value="trafficMonitoringStation" />
1607     <xs:enumeration value="transitOperator" />
1608     <xs:enumeration value="vehicleProbeMeasurement" />
1609     <xs:enumeration value="videoProcessingMonitoringStation" />
1610   </xs:restriction>
1611 </xs:simpleType>
1612 <xs:complexType name="SpeedManagement">
1613   <xs:complexContent>
1614     <xs:extension base="D2LogicalModel:NetworkManagement">
1615       <xs:sequence>
1616         <xs:element name="speedManagementType" type=
1617 ↪ "D2LogicalModel:SpeedManagementTypeEnum" minOccurs="0" maxOccurs="1" />
1618         <xs:element name="temporarySpeedLimit" type=
1619 ↪ "D2LogicalModel:KilometresPerHour" minOccurs="0" maxOccurs="1" />
1620         <xs:element name="speedManagementExtension" type="D2LogicalModel:_
1621 ↪ ExtensionType" minOccurs="0" />
1622       </xs:sequence>
1623     </xs:extension>
1624   </xs:complexContent>
1625 </xs:complexType>
1626 <xs:simpleType name="SpeedManagementTypeEnum">
1627   <xs:restriction base="xs:string">
1628     <xs:enumeration value="activeSpeedControlInOperation" />
1629     <xs:enumeration value="doNotSlowdownUnnecessarily" />
1630     <xs:enumeration value="observeSpeedLimit" />
1631     <xs:enumeration value="policeSpeedChecksInOperation" />
1632     <xs:enumeration value="reduceYourSpeed" />
1633     <xs:enumeration value="speedRestrictionInOperation" />
1634     <xs:enumeration value="other" />
1635   </xs:restriction>
1636 </xs:simpleType>
1637 <xs:simpleType name="String">
1638   <xs:restriction base="xs:string">
1639     <xs:maxLength value="1024" />
1640   </xs:restriction>
1641 </xs:simpleType>
1642 <xs:complexType name="Subjects">
1643   <xs:sequence>
1644     <xs:element name="subjectTypeOfWorks" type=
1645 ↪ "D2LogicalModel:SubjectTypeOfWorksEnum" minOccurs="1" maxOccurs="1" />
1646     <xs:element name="numberOfSubjects" type="D2LogicalModel:NonNegativeInteger"
1647 ↪ minOccurs="0" maxOccurs="1" />
1648     <xs:element name="subjectsExtension" type="D2LogicalModel:_ExtensionType"
1649 ↪ minOccurs="0" />
1650   </xs:sequence>
1651 </xs:complexType>

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1646 <xs:simpleType name="SubjectTypeOfWorksEnum">
1647   <xs:restriction base="xs:string">
1648     <xs:enumeration value="bridge" />
1649     <xs:enumeration value="buriedCables" />
1650     <xs:enumeration value="buriedServices" />
1651     <xs:enumeration value="crashBarrier" />
1652     <xs:enumeration value="gallery" />
1653     <xs:enumeration value="gantry" />
1654     <xs:enumeration value="gasMainWork" />
1655     <xs:enumeration value="interchange" />
1656     <xs:enumeration value="junction" />
1657     <xs:enumeration value="levelCrossing" />
1658     <xs:enumeration value="lightingSystem" />
1659     <xs:enumeration value="measurementEquipment" />
1660     <xs:enumeration value="noiseProtection" />
1661     <xs:enumeration value="road" />
1662     <xs:enumeration value="roadsideDrains" />
1663     <xs:enumeration value="roadsideEmbankment" />
1664     <xs:enumeration value="roadsideEquipment" />
1665     <xs:enumeration value="roadSigns" />
1666     <xs:enumeration value="roundabout" />
1667     <xs:enumeration value="tollGate" />
1668     <xs:enumeration value="tunnel" />
1669     <xs:enumeration value="waterMain" />
1670     <xs:enumeration value="other" />
1671   </xs:restriction>
1672 </xs:simpleType>
1673 <xs:complexType name="SupplementaryPositionalDescription">
1674   <xs:sequence>
1675     <xs:element name="locationDescriptor" type=
1676     ↪ "D2LogicalModel:LocationDescriptorEnum" minOccurs="0" maxOccurs="unbounded" />
1677     <xs:element name="sequentialRampNumber" type="D2LogicalModel:NonNegativeInteger
1678     ↪ " minOccurs="0" maxOccurs="1" />
1679     <xs:element name="affectedCarriagewayAndLanes" type=
1680     ↪ "D2LogicalModel:AffectedCarriagewayAndLanes" minOccurs="0" maxOccurs="unbounded" />
1681     <xs:element name="supplementaryPositionalDescriptionExtension" type=
1682     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1683   </xs:sequence>
1684   <xs:attribute name="locationPrecision" type=
1685   ↪ "D2LogicalModel:MetresAsNonNegativeInteger" use="optional" />
1686 </xs:complexType>
1687 <xs:simpleType name="Tonnes">
1688   <xs:restriction base="D2LogicalModel:Float" />
1689 </xs:simpleType>
1690 <xs:simpleType name="TrafficConstrictionTypeEnum">
1691   <xs:restriction base="xs:string">
1692     <xs:enumeration value="carriagewayBlocked" />
1693     <xs:enumeration value="carriagewayPartiallyObstructed" />
1694     <xs:enumeration value="lanesBlocked" />
1695     <xs:enumeration value="lanesPartiallyObstructed" />
1696     <xs:enumeration value="roadBlocked" />
1697     <xs:enumeration value="roadPartiallyObstructed" />
1698   </xs:restriction>
1699 </xs:simpleType>
1700 <xs:simpleType name="TrafficStatusEnum">
1701   <xs:restriction base="xs:string">

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1697     <xs:enumeration value="impossible" />
1698     <xs:enumeration value="congested" />
1699     <xs:enumeration value="heavy" />
1700     <xs:enumeration value="freeFlow" />
1701     <xs:enumeration value="unknown" />
1702 </xs:restriction>
1703 </xs:simpleType>
1704 <xs:complexType name="TrafficStatusValue">
1705   <xs:complexContent>
1706     <xs:extension base="D2LogicalModel:DataValue">
1707       <xs:sequence>
1708         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
1709 ↪ " minOccurs="1" maxOccurs="1" />
1710         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
1711 ↪ ExtensionType" minOccurs="0" />
1712       </xs:sequence>
1713     </xs:extension>
1714   </xs:complexContent>
1715 </xs:complexType>
1716 <xs:simpleType name="TrafficTypeEnum">
1717   <xs:restriction base="xs:string">
1718     <xs:enumeration value="accessOnlyTraffic" />
1719     <xs:enumeration value="destinedForAirport" />
1720     <xs:enumeration value="destinedForAirportArrivals" />
1721     <xs:enumeration value="destinedForAirportDepartures" />
1722     <xs:enumeration value="destinedForFerryService" />
1723     <xs:enumeration value="destinedForRailService" />
1724     <xs:enumeration value="holidayTraffic" />
1725     <xs:enumeration value="localTraffic" />
1726     <xs:enumeration value="longDistanceTraffic" />
1727     <xs:enumeration value="regionalTraffic" />
1728     <xs:enumeration value="residentsOnlyTraffic" />
1729     <xs:enumeration value="throughTraffic" />
1730     <xs:enumeration value="visitorTraffic" />
1731   </xs:restriction>
1732 </xs:simpleType>
1733 <xs:simpleType name="UrgencyEnum">
1734   <xs:restriction base="xs:string">
1735     <xs:enumeration value="extremelyUrgent" />
1736     <xs:enumeration value="urgent" />
1737     <xs:enumeration value="normalUrgency" />
1738   </xs:restriction>
1739 </xs:simpleType>
1740 <xs:simpleType name="Url">
1741   <xs:restriction base="xs:anyURI" />
1742 </xs:simpleType>
1743 <xs:complexType name="UrlLink">
1744   <xs:sequence>
1745     <xs:element name="urlLinkAddress" type="D2LogicalModel:Url" minOccurs="1"
1746 ↪ maxOccurs="1" />
1747     <xs:element name="urlLinkDescription" type="D2LogicalModel:MultilingualString"
1748 ↪ minOccurs="0" maxOccurs="1" />
1749     <xs:element name="urlLinkType" type="D2LogicalModel:UrlLinkTypeEnum" minOccurs=
1750 ↪ "0" maxOccurs="1" />
1751     <xs:element name="urlLinkExtension" type="D2LogicalModel:_ExtensionType"
1752 ↪ minOccurs="0" />

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1747     </xs:sequence>
1748 </xs:complexType>
1749 <xs:simpleType name="UrlLinkTypeEnum">
1750   <xs:restriction base="xs:string">
1751     <xs:enumeration value="documentPdf" />
1752     <xs:enumeration value="html" />
1753     <xs:enumeration value="image" />
1754     <xs:enumeration value="rss" />
1755     <xs:enumeration value="videoStream" />
1756     <xs:enumeration value="voiceStream" />
1757     <xs:enumeration value="other" />
1758   </xs:restriction>
1759 </xs:simpleType>
1760 <xs:complexType name="Validity">
1761   <xs:sequence>
1762     <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"
1763     ↪ minOccurs="1" maxOccurs="1" />
1764     <xs:element name="validityTimeSpecification" type="D2LogicalModel:OverallPeriod
1765     ↪ " />
1766     <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"
1767     ↪ minOccurs="0" />
1768   </xs:sequence>
1769 </xs:complexType>
1770 <xs:simpleType name="ValidityStatusEnum">
1771   <xs:restriction base="xs:string">
1772     <xs:enumeration value="active" />
1773     <xs:enumeration value="suspended" />
1774     <xs:enumeration value="definedByValidityTimeSpec" />
1775   </xs:restriction>
1776 </xs:simpleType>
1777 <xs:complexType name="VehicleCharacteristics">
1778   <xs:sequence>
1779     <xs:element name="loadType" type="D2LogicalModel:LoadTypeEnum" minOccurs="0"
1780     ↪ maxOccurs="1" />
1781     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
1782     ↪ "0" maxOccurs="unbounded" />
1783     <xs:element name="vehicleUsage" type="D2LogicalModel:VehicleUsageEnum"
1784     ↪ minOccurs="0" maxOccurs="1" />
1785     <xs:element name="grossWeightCharacteristic" type=
1786     ↪ "D2LogicalModel:GrossWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1787     <xs:element name="heightCharacteristic" type=
1788     ↪ "D2LogicalModel:HeightCharacteristic" minOccurs="0" maxOccurs="2" />
1789     <xs:element name="lengthCharacteristic" type=
1790     ↪ "D2LogicalModel:LengthCharacteristic" minOccurs="0" maxOccurs="2" />
1791     <xs:element name="widthCharacteristic" type="D2LogicalModel:WidthCharacteristic
1792     ↪ " minOccurs="0" maxOccurs="2" />
1793     <xs:element name="heaviestAxleWeightCharacteristic" type=
1794     ↪ "D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1795     <xs:element name="numberOfAxlesCharacteristic" type=
1796     ↪ "D2LogicalModel:NumberOfAxlesCharacteristic" minOccurs="0" maxOccurs="2" />
1797     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
1798     ↪ ExtensionType" minOccurs="0" />
1799   </xs:sequence>
1800 </xs:complexType>
1801 <xs:complexType name="VehicleCountValue">
1802   <xs:complexContent>

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1790     <xs:extension base="D2LogicalModel:DataValue">
1791         <xs:sequence>
1792             <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"
1793             ↪ minOccurs="1" maxOccurs="1" />
1794             <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
1795             ↪ ExtensionType" minOccurs="0" />
1796         </xs:sequence>
1797     </xs:extension>
1798 </xs:complexContent>
1799 </xs:complexType>
1800 <xs:complexType name="VehicleFlowValue">
1801     <xs:complexContent>
1802         <xs:extension base="D2LogicalModel:DataValue">
1803             <xs:sequence>
1804                 <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"
1805                 ↪ minOccurs="1" maxOccurs="1" />
1806                 <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
1807                 ↪ ExtensionType" minOccurs="0" />
1808             </xs:sequence>
1809         </xs:extension>
1810     </xs:complexContent>
1811 </xs:complexType>
1812 <xs:simpleType name="VehiclesPerHour">
1813     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1814 </xs:simpleType>
1815 <xs:simpleType name="VehicleTypeEnum">
1816     <xs:restriction base="xs:string">
1817         <xs:enumeration value="agriculturalVehicle" />
1818         <xs:enumeration value="anyVehicle" />
1819         <xs:enumeration value="articulatedVehicle" />
1820         <xs:enumeration value="bicycle" />
1821         <xs:enumeration value="bus" />
1822         <xs:enumeration value="car" />
1823         <xs:enumeration value="caravan" />
1824         <xs:enumeration value="carOrLightVehicle" />
1825         <xs:enumeration value="carWithCaravan" />
1826         <xs:enumeration value="carWithTrailer" />
1827         <xs:enumeration value="constructionOrMaintenanceVehicle" />
1828         <xs:enumeration value="fourWheelDrive" />
1829         <xs:enumeration value="highSidedVehicle" />
1830         <xs:enumeration value="lorry" />
1831         <xs:enumeration value="moped" />
1832         <xs:enumeration value="motorcycle" />
1833         <xs:enumeration value="motorcycleWithSideCar" />
1834         <xs:enumeration value="motorscooter" />
1835         <xs:enumeration value="tanker" />
1836         <xs:enumeration value="threeWheeledVehicle" />
1837         <xs:enumeration value="trailer" />
1838         <xs:enumeration value="tram" />
1839         <xs:enumeration value="twoWheeledVehicle" />
1840         <xs:enumeration value="van" />
1841         <xs:enumeration value="vehicleWithCatalyticConverter" />
1842         <xs:enumeration value="vehicleWithoutCatalyticConverter" />
1843         <xs:enumeration value="vehicleWithCaravan" />
1844         <xs:enumeration value="vehicleWithTrailer" />
1845         <xs:enumeration value="withEvenNumberedRegistrationPlates" />

```

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1842     <xs:enumeration value="withOddNumberedRegistrationPlates" />
1843     <xs:enumeration value="other" />
1844   </xs:restriction>
1845 </xs:simpleType>
1846 <xs:simpleType name="VehicleUsageEnum">
1847   <xs:restriction base="xs:string">
1848     <xs:enumeration value="agricultural" />
1849     <xs:enumeration value="commercial" />
1850     <xs:enumeration value="emergencyServices" />
1851     <xs:enumeration value="military" />
1852     <xs:enumeration value="nonCommercial" />
1853     <xs:enumeration value="patrol" />
1854     <xs:enumeration value="recoveryServices" />
1855     <xs:enumeration value="roadMaintenanceOrConstruction" />
1856     <xs:enumeration value="roadOperator" />
1857     <xs:enumeration value="taxi" />
1858   </xs:restriction>
1859 </xs:simpleType>
1860 <xs:complexType name="WidthCharacteristic">
1861   <xs:sequence>
1862     <xs:element name="comparisonOperator" type=
1863     ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1864     <xs:element name="vehicleWidth" type="D2LogicalModel:MetresAsFloat" minOccurs="1"
1865     ↪ " maxOccurs="1" />
1866     <xs:element name="widthCharacteristicExtension" type="D2LogicalModel:_
1867     ↪ ExtensionType" minOccurs="0" />
1868   </xs:sequence>
1869 </xs:complexType>
1870 <xs:complexType name="WinterDrivingManagement">
1871   <xs:complexContent>
1872     <xs:extension base="D2LogicalModel:NetworkManagement">
1873       <xs:sequence>
1874         <xs:element name="winterEquipmentManagementType" type=
1875         ↪ "D2LogicalModel:WinterEquipmentManagementTypeEnum" minOccurs="1" maxOccurs="1" />
1876         <xs:element name="winterDrivingManagementExtension" type="D2LogicalModel:_
1877         ↪ ExtensionType" minOccurs="0" />
1878       </xs:sequence>
1879     </xs:extension>
1880   </xs:complexContent>
1881 </xs:complexType>
1882 <xs:simpleType name="WinterEquipmentManagementTypeEnum">
1883   <xs:restriction base="xs:string">
1884     <xs:enumeration value="doNoUseStudTyres" />
1885     <xs:enumeration value="useSnowChains" />
1886     <xs:enumeration value="useSnowChainsOrTyres" />
1887     <xs:enumeration value="useSnowTyres" />
1888     <xs:enumeration value="winterEquipmentOnBoardRequired" />
1889     <xs:enumeration value="other" />
1890   </xs:restriction>
1891 </xs:simpleType>
1892 </xs:schema>

```

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- [16157-5] CEN TS 16157-5: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 5: Measured and elaborated data publications
- [d2ref] DATEX II Current version/Reference set, <https://docs.datex2.eu/downloads/>.
- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.

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